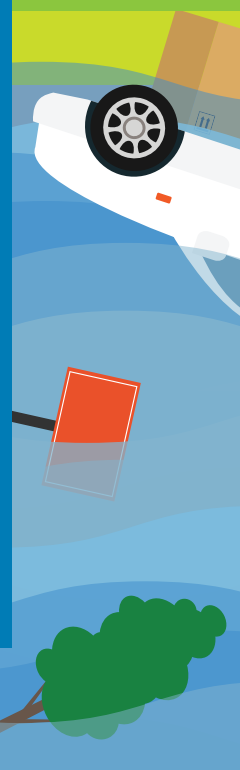




THE CLIMATE-RESILIENT ENERGY IMPERATIVE

ADAPTATION BENEFITS AND ACTIONS FOR ALL

JULY 2026



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Notes:

1. In this publication, “\$” refers to United States dollars.
2. ADB recognizes “Laos” as the Lao People’s Democratic Republic and “Republic of Marshall Islands” as the Republic of the Marshall Islands.
3. Effective 1 February 2021, ADB place a temporary hold on sovereign project disbursements and new contracts in Myanmar ADB placed its regular assistance to Afghanistan on hold, effective 15 August 2021.

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Abbreviations

ADB	Asian Development Bank
AI	artificial intelligence
CPS	country partnership strategy
DMC	developing member country
ENSO	El Niño Southern Oscillation
IEA	International Energy Agency
IPCC	Intergovernmental Panel on Climate Change
IRENA	International Renewable Energy Agency
Lao PDR	Lao People's Democratic Republic
MDB	multilateral development bank
NAP	national adaptation plan
NDC	nationally determined contribution
OCR	ordinary capital resources
PPP	public-private partnership
PRC	People's Republic of China
PV	photovoltaic
UNFCCC	United Nations Framework Convention on Climate Change
VRE	variable renewable energy
WMO	World Meteorological Organization

Executive Summary

Climate-resilient energy systems are an imperative now. Rising temperatures, shifting precipitation patterns, sea level rise, and more frequent extreme weather events are disrupting energy supply while reshaping energy demand patterns. Energy systems in Asia and the Pacific are already struggling to cope with power outages caused by extreme heat events, physical damage to assets by floods and landslides, increasing threats to coastal energy infrastructure due to tropical cyclones and sea level rise, and changes in energy resource availability. As the energy sector in the region is already strained by weak infrastructure and rapidly increasing demand, such impacts due to the changing climate pose a significant challenge to energy system resilience.

Recognizing these growing risks, diverse stakeholders—including energy suppliers, consumers, and national authorities—are increasingly calling for stronger resilience measures. Energy suppliers seek to protect assets, maintain service reliability, and safeguard revenue streams from disruptions due to changes in the climate and extreme weather. Resilient energy systems help energy suppliers avoid costly damage, pursue long-term planning, and strengthen stability. For consumers, reliable energy access underpins economic growth, ensures the continuity of essential public services such as health care and water supply, and safeguards livelihoods—especially for vulnerable groups who lack alternative resources to cope with disruptions. Governments and regulators view energy resilience as essential for ensuring energy security, advancing renewable energy transitions, and enhancing disaster risk management capacity.

Despite rising awareness and clear benefits, adaptation in the energy sector lags behind. Many efforts focus narrowly on individual assets rather than developing system-wide, long-term resilience. The report identifies four critical barriers to progress:

- **Limited availability of accessible and accurate climate and weather information for energy.** Although climate and weather data are becoming increasingly important with growing impacts of the changing climate and the rapid expansion of renewable energy, developers and investors frequently report challenges in accessing and interpreting climate information. Government support also remains limited despite its critical role in the collection and dissemination of such information.
- **Limited consideration of energy system's adaptation in national climate strategies.** Discussions on energy and climate have traditionally concentrated on mitigation while adaptation has received significantly less attention. This imbalance is particularly evident in national climate strategies such as nationally determined contributions (NDCs) and national adaptation plans (NAPs). Globally, only 41% of NDCs submitted by 2024 prioritized adaptation in the energy sector, while only 27% of global NAPs (13 out of 48, as of 2023) recognized energy as an adaptation priority. In Asia and the Pacific, 43% of NDCs and 63% of NAPs (as of October 2025) included energy as one of their adaptation priorities. Although these efforts align with, or even exceed, global trends, they remain insufficient given the scale of the impact of the changing climate on energy systems.

- **Insufficient adaptation finance in the energy sector.** Although energy is among the sectors requiring the largest amount of adaptation financing, it continues to fall short as most climate finance goes to mitigation. For instance, in 2023, MDBs committed only \$7.25 billion for adaptation in energy, transport, and other built environment and infrastructure, accounting for less than 10% of total climate finance committed to these sectors (\$76.45 billion). Moreover, adaptation finance from the private sector remains limited, particularly in developing countries, due to perceived high up-front costs, overlooked long-term adaptation benefits, lack of reliable climate risk data, monopolistic market structures, and policy uncertainties.
- **Slow progress in adopting innovative technologies.** In the energy sector, the uptake of innovative adaptation technologies has lagged even though many of these technologies are both mature and economically viable. Variable renewable energy technologies, which enable diversification and decentralization of energy sources against climate impacts, remain limited in many developing countries. Deployment of smart grids and efficient cooling solutions also remains low. The slow progress is due to various reasons including low awareness, limited technical capacity, high financing costs, weak institutional frameworks, and the need to adapt technologies to local contexts.

To overcome these barriers and strengthen resilience, the following actions from all key stakeholders, including energy suppliers, consumers and authorities, are essential (Table):

- **Improve quality of climate information for energy systems.** Government authorities, including relevant ministries and agencies, can address knowledge barriers by enhancing the quality of climate information services and strengthening observation networks near major energy assets. Energy suppliers can contribute to improving climate information by conducting robust climate risk and impact assessments and feeding the results back into national information systems. Energy consumers can also provide data on climate impacts from the demand side through monitoring and feedback systems.
- **Incorporate climate-resilient energy systems in national strategies.** National authorities can avoid market failures and enforce resilience measures to protect broader socioeconomic benefits by recognizing the energy sector as an adaptation priority. As of October 2025, 22 out of 42 the Asian Development Bank developing member countries included the energy sector as an adaptation priority in their NDCs or NAPs. Energy suppliers may be encouraged to embed resilience considerations into project design and operations, following the recommendations outlined in national climate strategies. Energy consumers can advocate for climate resilience, raise public awareness of climate risks, and foster broader acceptance of adaptation measures.
- **Scale up adaptation finance in the energy sector.** Private energy suppliers and investors can expand adaptation finance by embedding climate risk assessments and resilience benefits into their decision-making processes. This approach enables them to better understand the financial impacts of climate hazards and enhance long-term revenues. Public investment from national authorities and multilateral development banks can help lower costs, boost investor confidence, and facilitate the implementation of resilience measures. Energy consumers can encourage energy suppliers to adopt resilience measures through power purchase agreements that include reliability clauses against climate-driven disruptions.

Table: Actions to Build Climate Resilience in the Energy Sector by Stakeholder

STAKEHOLDERS	National authorities	Energy suppliers	Energy consumers
ACTIONS			
Actions for Improved Quality of Climate Information for Energy Systems by Stakeholder 	- Enhance the quality of public climate information services at national, regional, and local levels - Strengthen climate observation networks by expanding observation stations near major energy assets and introducing advanced technologies	- Conduct robust climate risk and impact assessments - Feed assessment results back into national climate information systems	- Offer valuable feedback on how climate and weather influence their energy consumption patterns and comfort levels - Install energy consumption monitoring systems to enable more detailed analysis of climate variables and energy demand patterns
Actions for Incorporation of Climate-resilient Energy Systems in National Strategies by Stakeholder 	- Include the energy sector in national adaptation strategies as one of the adaptation priorities - Organize stakeholder consultations to identify concrete actions across the entire lifecycle of energy projects - Develop a robust implementation plan	- Embed resilience considerations into project design and operations - Align corporate strategies with national adaptation goals - Provide technical feedback and share best practices for further development and effective implementation of adaptation policies	- Provide a strong political foundation for advancing climate resilience policies and regulations - Raise public awareness of climate risks to the energy sector - Foster broader acceptance of adaptation measures
Actions for Scaling Up Adaptation Finance in the Energy Sector by Stakeholder 	- Promote blended finance approaches that combine concessional capital from the public sector with private sector investments - Create an enabling environment for private investment by providing technical assistance for climate risk assessments and demonstrating the viability of innovative technologies - Offer risk-sharing mechanisms through diverse and innovative financial instruments to scale up adaptation finance	- Expand adaptation finance by systematically embedding climate risk assessments in investment planning - Incorporate long-term resilience benefits into decision-making and valuation processes - Leverage insurance as an effective risk-transfer mechanism to manage climate-related uncertainties	Integrate reliability requirements into power purchase agreements
Actions for Accelerating Deployment of Innovative Adaptation Technologies in the Energy Sector by Stakeholder 	- Support the adoption of innovative energy-efficient technologies by improving standards and labeling systems - Encourage behavioral changes by providing incentives	- Scale up new renewable energy technologies for diversification of energy sources - Improve the accuracy of forecasts of potential outages and enable early warning systems through digital technologies - Enable faster detection and restoration of climate-related faults with advanced smart technologies	- Enhance the accuracy and applicability of digital technologies as a knowledge provider - Accelerate the deployment of innovative technologies by purchasing more efficient options

- **Accelerate deployment of innovative adaptation technologies.** Energy suppliers can enhance resilience through technological and geographical diversification of energy sources. The increasing penetration of new renewable energy can be a resilience solution when its climate sensitivity is addressed through innovative designs and technologies. Digital technologies also enhance resilience in the energy sector by providing tools to monitor, mitigate, respond to, and recover from climate-driven disruptions. Energy consumers can support the deployment of innovative adaptation solutions by acting as knowledge providers for digital technologies and choosing more efficient energy options. Governments can support the adoption of innovative energy-efficient technologies by improving standards and labeling systems and encouraging behavioral changes through incentives.

Building climate-resilient energy systems is no longer a choice but an urgent necessity for Asia and the Pacific. It is a shared responsibility that calls for immediate and coordinated action from all stakeholders. National authorities play a central role in enhancing the quality of climate information for energy systems and embedding resilience into national energy and climate strategies. Energy suppliers take the lead in scaling up adaptation finance and advancing the deployment of innovative adaptation technologies. Consumers contribute by providing feedback and information, fostering political support, and advocating for greater reliability and efficiency in energy services. Collective action grounded in clearly defined roles, accountability, and shared commitment can transform the region's energy systems into a foundation for resilience and prosperity across Asia and the Pacific.

1 Introduction

Increasing climate variability is posing unprecedented challenges to global energy systems, impacting the entire energy value chain, including fuel supply, electricity generation, transmission, and distribution, as well as energy demand. Climate impacts test energy infrastructure resilience, change generation potential, reduce efficiency, and alter demand patterns, affecting energy security and sustainable development. In many developing countries in Asia and the Pacific, climate risks have become a significant concern for energy systems that are already strained by rising energy demand, limited financial resources, and aging infrastructure.

Despite the growing urgency to enhance resilience of energy systems, adaptation actions remain critically insufficient, leaving energy infrastructure vulnerable to escalating climate impacts. Until now, most adaptation actions in the energy sector have been concentrated on climate-proofing of infrastructure against immediate threats rather than building long-term resilience of energy systems despite its multiple benefits. Limited awareness of long-term adaptation needs for energy has led to the sector's being overlooked in national adaptation priorities, despite the tremendous benefits of resilient energy systems.

This report aims to increase awareness of climate impacts on the energy sector and identify adaptation measures for key stakeholders to build resilient energy systems. It provides a snapshot of climate impacts on energy and discusses the multiple benefits of building resilience. It identifies existing barriers and proposes actions for crucial actors including energy suppliers, consumers, and government and regulators to bridge the gap.

The report consists of six chapters. Following Chapter 1's introduction, Chapter 2 explains why the energy sector requires urgent adaptation measures and discusses climate impacts on energy infrastructure. Chapter 3 focuses on the benefits of resilient energy systems, highlighting the growing demand for building resilience from diverse stakeholders, including energy suppliers, consumers, and authorities. Chapter 4 identifies key barriers to adaptation in terms of information, policy, finance, and technologies. Chapter 5 explores strategies for overcoming these barriers and suggests specific actions for each key stakeholder, followed by Chapter 6's conclusions.

2 Why Should Energy Systems Be Resilient to Climate Impacts?

In recent years increasing impacts due to the changes in the climate have highlighted the growing urgency for building resilient energy systems. The record-breaking global warming in 2024, 1.55°C above pre-industrial levels, has triggered severe and widespread impacts, which significantly disrupted energy supply chains, challenged infrastructure resilience, and reshaped energy demand patterns. The situation is particularly critical in Asia and the Pacific, a region experiencing faster-than-global-average temperature increases and notable disruptions to its already vulnerable energy systems. Energy systems in Asia and the Pacific have already experienced energy supply disruptions, infrastructure failures, and increasing energy consumption. Without immediate actions to enhance the resilience of energy systems through investments in adaptation, increasing risks could threaten sustainable development in the region.

Increasing Climate Impacts

In recent years impacts due to changes in the climate have intensified. The year 2024 was the warmest year on record, reaching the global mean surface temperature of 1.55 ± 0.13 °C above the pre-industrial level (1850–1900), making it the first year with a global mean temperature rise of more than 1.5°C.¹ The record-breaking warming is alarming since it is notably higher than the levels of previous warmest years (2016 at 1.29°C, 2020 at 1.27°C and 2023 at 1.45°C above the pre-industrial level).² The unprecedented warmth triggered other impacts such as a record-high sea level rise and glacier retreat. The rising temperatures also intensified extreme weather events such as heatwaves, floods, droughts, wildfires, and tropical cyclones across the world, bringing devastating socioeconomic impacts to many countries.

Asia and the Pacific is the most exposed region to climate-related hazards in the world.³ 2024 was the second warmest year in Asia, while sea surface temperatures in the Indian Ocean and the western Pacific Ocean marked a record high.⁴ The World Meteorological Organization (WMO) reported that the Asia and Pacific region experienced a faster increase in temperatures than the world average in 1991–2024, and the warming exacerbated.⁵

¹ World Meteorological Organization (WMO). 2025. [WMO Confirms 2024 as Warmest Year On Record at About 1.55°C Above Pre-industrial Level](#).

² WMO. 2024. [State of the Global Climate 2023](#).

³ World Bank. 2024. [Rising to the Challenge](#).

⁴ National Oceanic and Atmospheric Administration (NOAA). 2025. [2024 Was the World's Warmest Year on Record](#); and Copernicus. 2025. [Copernicus: 2024 Is the First Year to Exceed 1.5°C Above Pre-industrial Level](#).

⁵ WMO. 2025. [State of the Climate in Asia 2024](#).

Indeed, in March–April 2024, multiple Southeast Asian countries experienced heatwaves which resulted in 61 deaths in Thailand⁶ and school shutdowns in the Philippines and Cambodia.⁷ Several parts of India experienced intense heatwaves which caused more than 450 deaths across the country.⁸ North Pacific countries such as the Federated States of Micronesia and the Republic of the Marshall Islands experienced a severe drought in March 2024. Over 26,000 people were affected by water shortages, which continued for several months.⁹ On the other hand, other parts of Asia and the Pacific experienced extreme precipitation events. In Central Asia, record-high extreme rainfall in March led to flooding across large areas, while some parts of India and Pakistan experienced heavy rainfall, floods, and landslides as well.¹⁰

Climate Impacts on Energy

Energy is one of the sectors witnessing increasing pressure from climate impacts. The Intergovernmental Panel on Climate Change (IPCC) recognizes that many components of the energy system are affected not only by individual extreme weather events and climate conditions but also by a range of compounding effects of multiple weather and climate conditions (Table 1).¹¹

Energy resource supply can be disrupted by extreme weather events and their compounding effects. For example, floods and droughts can negatively affect resource supply chains, including biofuel production, critical minerals mining, and coal processing. In some regions, the combination of hot and dry conditions intensifies water stress, heightening competition for water supply between energy production and other sectors. Moreover, tropical cyclones when coupled with sea level rise can disrupt major coastal fuel production and storage facilities through storm surges, coastal flooding, and erosion.

Electricity supply can be also interrupted by climate impacts which may shift electricity generation potential and efficiency. For example, increasing frequencies of floods and droughts necessitate design changes in hydropower plants, including reservoir sizes, dam heights, canals, and turbines, as well as revisions to operating regimes and emergency response plans. Extreme heat events heighten the need for cooling solutions for solar photovoltaic systems, which are sensitive to temperatures above 25°C. Rising water temperatures and declining water availability in some regions pose additional challenges for thermal power plants that rely on water for cooling. Similarly, intensifying tropical cyclones and decreasing average wind speeds in certain areas can reduce power generation from wind turbines, which operate effectively only within specific wind speed ranges.

More frequent extreme weather events test the physical resilience of electricity transmission and distribution networks, potentially leading to widespread system failures (Box 1). Electricity networks form the backbone of reliable and affordable electricity systems, yet they are also a major source of climate-induced outages and are considered the most vulnerable component of the electricity value chain. Tropical cyclones can damage power lines, poles, and transformers by toppling trees and branches. Rising temperatures may cause failures as power lines expand and sag. The combined effects of rising temperatures and droughts can trigger wildfires, which in turn cause simultaneous multiple faults in the grid. Even underground cables, which are generally more resilient than overhead lines, can experience faults as the combined effects of heat and drought place particular stress on insulating materials.

⁶ The Straits Times. 2024. *Heatstroke Kills 61 in Thailand So Far In 2024*. 10 May.

⁷ United Nations Office for Disaster Risk Reduction (UNDRR). 2024. *Intense Heatwaves Make April 2024 the Hottest On Record*.

⁸ Footnote 5.

⁹ Reliefweb. 2024. *UNICEF Pacific: Drought Emergency Response for the North Pacific – Federated States of Micronesia and Republic of Marshall Islands*.

¹⁰ Footnote 5.

¹¹ IPCC. 2021. *IPCC Sixth Assessment Report: Working Group III: Mitigation of Climate Change*.

Table 1: Climate Risks to Electricity Supply in the Low-emissions and High-emissions Scenarios, 2080–2100

Low-emission scenario								
	Coal	Gas	Oil	Nuclear	Hydro	Solar	Wind	Grid
Temperature	●	●	●	●	●	●	●	●
Precipitation (dry)	●	●	●	●	●	●	●	●
Precipitation (wet)	●	●	●	●	●	●	●	●
Sea level	●	●	●	●	●	●	●	●
Wind (cyclones)	●	●	●	●	●	●	●	●
Wildfires	●	●	●	●	●	●	●	●

High-emission scenario								
	Coal	Gas	Oil	Nuclear	Hydro	Solar	Wind	Grid
Temperature	●	●	●	●	●	●	●	●
Precipitation (dry)	●	●	●	●	●	●	●	●
Precipitation (wet)	●	●	●	●	●	●	●	●
Sea level	●	●	●	●	●	●	●	●
Wind (cyclones)	●	●	●	●	●	●	●	●
Wildfires	●	●	●	●	●	●	●	●

Notes: These tables show climate risks to power plants and electricity grids at a global scale. SSP1–2.6 and SSP5–8.5 are used for the low-emissions and high-emissions scenarios, respectively. The levels of climate risks are redivided into five categories, from dark green for low risks to red for high risks. Gray cells indicate no information. The levels are determined based on the combination of hazard, exposure, and vulnerability. Hazard and exposure are calculated through geographic information system analysis while vulnerability is determined through qualitative research. For temperature, the level of exposure of power plants and grids to the increase in mean temperature and the number of days with maximum temperature above 35°C data is used. For precipitation (dry), the level of exposure of power plants and grids to a drier climate based on the Standard Precipitation Index and the number of consecutive dry days is used. For precipitation (wet), the level of exposure of power plants and grids to a wetter climate based on the Standard Precipitation Index and 1-day maximum precipitation is used. For sea level, the level of exposure of power plants and grids to the projected rise of sea level is used. For wind, the level of exposure of power plants and grids to tropical cyclones based on the analysis of historical trends of tropical cyclones and major tropical cyclones (above Category 3) is used. For wildfires, the level of exposure of power plants and grids to wildfires based on the analysis of historical trends of fire weather is used.

Source: International Energy Agency. 2022. *Climate Resilience for Energy Security*.

Box 1: Border-Crossing Climate Impacts Through Interconnected Electricity Networks

Transboundary electricity networks that connect energy resources across multiple countries are expected to play an increasingly important role in stabilizing electricity supply and driving the integration of renewable sources. By linking diverse energy sources and aggregating multiple supply options, interconnected networks help countries strengthen collective energy security by pooling generation capacity across borders, balancing electricity flows, and providing backup power during domestic disruptions. Well-designed interconnections with adequate flexibility and redundancy allow regional systems to maintain electricity supply through alternative sources even when parts of the network experience climate-driven disruptions. They also reduce the risk of prolonged outages that can disrupt essential services and economic activity.

While interconnected networks offer many benefits, they can also introduce shared challenges if not carefully designed. When a transboundary grid depends heavily on limited energy sources or transmission routes, disruptions in one part of the system may have cascading impacts across the network. As climate impacts are projected to increase, a climate-driven disruption to one part of the interconnected network may compromise the reliability of the entire system.

To ensure that interconnected electricity systems enhance, rather than undermine, energy security, they need to be designed with resilience in mind. Geographic distribution of energy sources, diversification of energy technologies, the use of resilient materials, energy storage systems, greater redundancy, and the deployment of digital technologies for early warning and monitoring contribute to building resilient interconnected networks. By incorporating climate resilience into their design, transboundary electricity networks can significantly enhance regional energy security and serve as a foundation for reliable and sustainable energy systems.

Source: Author.

Changing climate patterns are reshaping energy demand worldwide. Rising temperatures drive higher demand for cooling while reducing the need for heating in most regions. In areas experiencing greater precipitation variability and more frequent droughts, electricity consumption for water supply tends to increase. When droughts are compounded by sea level rise in islands and coastal regions, saltwater intrusion accelerates, placing additional pressure on already stressed water supplies. In these cases, countries are forced to adopt more energy-intensive solutions, such as desalination and deeper well pumping, as surface water availability declines and groundwater levels drop.

The power sector is one of the most exposed infrastructure sectors to climate risks. It accounts for nearly 30% of global value of infrastructure exposed to climate risks and around 37% of average annual loss of infrastructure sectors (Figure 1). According to the average annual loss analysis, flood and wind are associated with around two-thirds of the power sector's loss.¹² It highlights the urgent need for undertaking resilience measures for power infrastructure.

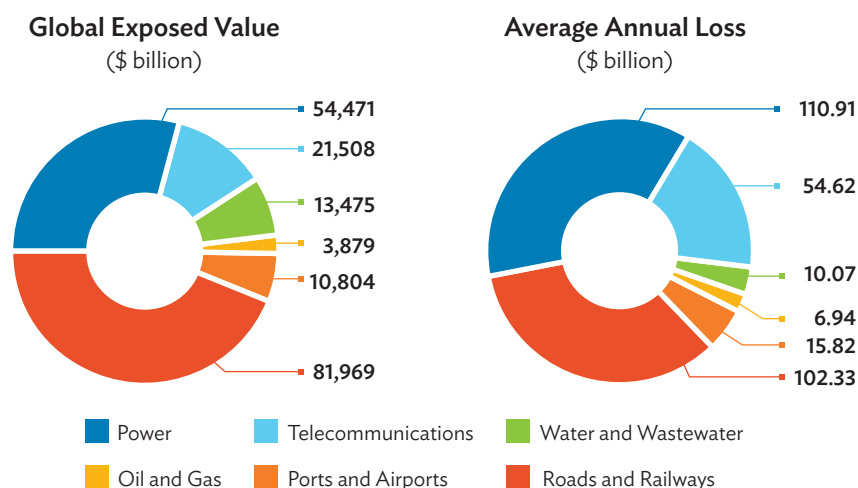
Energy systems in Asia and the Pacific are already suffering from climate impacts. Vientiane, the capital of the Lao People's Democratic Republic (Lao PDR), experienced multiple power outages in April–May 2024 due to a surge in electricity consumption and malfunctioning of overheated electrical systems amid heatwaves.¹³ In October 2024, electricity supply to over 970,000 customers was disrupted because of the damage from Tropical Storm Kristine (Tami) in the Philippines.¹⁴ A significant glacial lake outburst flood induced by glacier retreat led to the catastrophic collapse of the 1.2-gigawatt Teesta III hydroelectric dam in India in October 2023 causing widespread devastation including over 100 deaths and around 4,500 evacuations.¹⁵

¹² Coalition for Disaster Resilient Infrastructure. 2023. *Global Infrastructure Resilience Report*.

¹³ P. Visapra. 2024. *Heatwave Strains Power Supply Across Laos: Electricity State Company Responds to Public Backlash*. *The Lao Times*. 9 May.

¹⁴ K. Crismundo. 2024. *Over 970K Customers Affected by Outages Amid 'Kristine' Onslaught*. *Philippines News Agency*. 23 Oct.

¹⁵ World Meteorological Organization (WMO). 2024. *State of the Climate in Asia 2023*.

Figure 1: Exposed Value and Average Annual Loss by Infrastructure Sector

Source: Coalition for Disaster Resilient Infrastructure. 2023. *Global Infrastructure Resilience Report*.

Climate-driven disruptions in the energy sector are expected to intensify in the coming decades. Rising temperatures are accelerating glacier melt in the Hindu Kush Himalayan region, increasing the urgency for resilience measures in hydroelectric dams that depend on glacial runoff¹⁶ and demanding the diversification of power supply options. Flood risks will become higher in monsoon regions in South Asia, Southeast Asia, and East Asia, while drought risks increase in areas in West Asia, Central Asia, and South Asia.¹⁷ The escalation in seasonal and geographical variabilities in precipitation can have direct impacts on energy, altering water levels for hydropower generation and cooling thermal power plants. Additionally, rising sea levels and the increasing intensity of tropical cyclones pose severe risks to coastal energy infrastructure, including power plants, fuel production sites, and transmission networks, many of which are situated near shorelines.

Beyond these physical impacts on the energy supply side, a warmer climate is also altering energy demand. According to the IPCC Sixth Assessment Report, increasing temperatures and more frequent heatwaves will drive a net rise in global energy demand, even though reduced heating demand in winter may partially offset increased cooling needs in summer.¹⁸ The rising electricity demand can challenge energy security in developing countries in Asia where power supply is already strained due to population growth, economic development, and aging infrastructure. The Asian Development Bank (ADB) estimates that higher temperatures increase demand for cooling enough to cause a 3.3% GDP reduction for the Asia and Pacific region by 2070.¹⁹

¹⁶ International Energy Agency (IEA). 2021. *Climate Impacts on South and Southeast Asian Hydropower*.

¹⁷ IPCC. 2021. *IPCC Sixth Assessment Report: Working Group II: Impacts, Adaptation and Vulnerability*.

¹⁸ Footnote 17.

¹⁹ ADB. 2024. *Asia-Pacific Climate Report 2024: Catalyzing Finance and Policy Solutions*.

3 Who Will Benefit from Resilient Energy Systems?

As climate impacts intensify, the demand for resilient energy systems is increasing among a wide range of stakeholders, including energy suppliers, consumers, and government authorities. Energy suppliers aim to protect their assets to provide reliable service and thereby maintain revenue stream in the face of climate-related disruptions. Resilient energy systems also help energy suppliers prevent costly damage, while supporting long-term planning and enhancing supply chain stability. For consumers, reliable energy access underpins economic growth, safeguards public services like health care and water supply, and protects livelihoods. This is especially crucial for vulnerable groups who often lack the resources to cope with energy supply disruptions or find alternatives. Governments and regulators recognize the urgency of securing energy systems to achieve national energy security and development objectives. Ultimately, resilient energy systems not only reduce climate-related risks but also unlock broad economic and social benefits that far outweigh the up-front investment costs of resilience measures.

Growing Demand for Building Resilience

With intensifying climate impacts, diverse stakeholders increasingly demand resilient energy systems. The key actors include energy suppliers, consumers, and authorities which encompass national and subnational governments as well as regulators. These key actors collaborate with other actors such as academia, international organizations, and civil society.

Energy suppliers, such as generators and operators, bear primary responsibility for and have a direct interest in safeguarding their assets and offering reliable energy services for their customers. When climate-driven disruptions occur, suppliers incur revenue losses and repair expenses.

Energy consumers, also known as end users, ultimately bear the largest share of climate-driven disruption costs. Given the dependency of most economic and social sectors on stable energy supply, consumers face consequences such as interrupted critical social services (e.g., health care, education, and irrigation) and diminished economic productivity. Such impacts may disproportionately affect vulnerable populations who lack alternative energy options or financial means to mitigate costs (Box 2).

National authorities have a direct interest in maintaining a resilient energy supply to achieve their objectives for energy security while supporting broader socioeconomic goals, including protecting citizen welfare and managing disaster risks. Authorities thus play an essential role in establishing necessary policies and regulations, providing climate risk information, facilitating proactive actions by both energy suppliers and consumers, diversifying energy mix, and raising awareness about the comprehensive socioeconomic benefits associated with resilient energy systems.

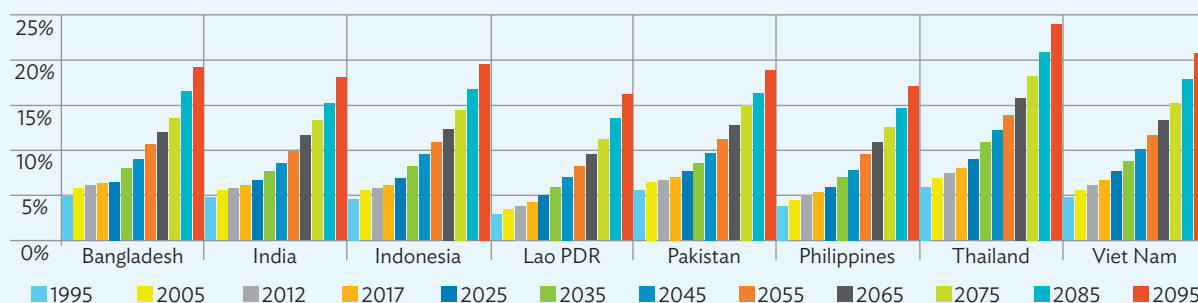
Box 2: Resilient Energy Systems for Vulnerable Groups

Resilient energy systems are crucial for Asia and the Pacific, one of the regions most vulnerable to climate hazards. Climate-driven disruptions in energy supply disproportionately affect vulnerable groups in this region, which is already grappling with widening income gaps, persistent gender inequalities, and a growing population of both young and elderly individuals.

Reliable and resilient energy services are essential for protecting poor people during extreme weather events. When extreme weather events cause energy disruptions, low-income households often lack the resources to access alternative energy sources or afford costly backup solutions. More likely to live in areas with older and weakened grid infrastructure, lower-income and marginalized communities tend to experience 2.5–4 times longer outage until restoration as targeted restorations may focus on areas of wealthier and more advantaged communities.^a To address the issue of energy injustice, energy storage technologies have been proposed as a solution to mitigate and prevent climate-driven disruptions and to enhance community resilience. For instance, the Asian Development Bank supported the installation of battery energy storage systems in the outer islands of Nukufetau, Nukulaelae, and Nui of Tuvalu to ensure resilience of electricity systems in marginalized communities.^b

In addition, extreme weather events such as heatwaves can disrupt the livelihoods of poor people, particularly those working under limited cooling options, leading to missed working hours and reduced productivity. Reliable electricity supply, which can enable cooling solutions during extreme heat events, helps workers maintain productivity and secure economic income.

Annual Work Output Lost Due to Heat (%)



Lao PDR = Lao People's Democratic Republic.

Note: The country average is based on population distribution and estimated heat levels in each grid cell under the Representative Concentration Pathway 6.0 scenario with 300 watts work intensity.

Source: ADB. 2022. *Beating the Heat: Investing in Pro-poor Solutions for Urban Resilience*.

Resilient energy systems can significantly advance gender equality in Asia and the Pacific, where structural inequalities heighten women's vulnerability to climate hazards. In many developing countries, women typically bear the responsibility for collecting water and fuel, tasks that become increasingly challenging during extreme weather events such as droughts and floods. Resilient energy systems that support cooking and water pumping can substantially reduce these burdens, freeing women's time and enabling greater access to education and training.

A stable energy supply is also vital for protecting children and elderly individuals who are particularly susceptible to heat-related illnesses. Extreme heat can cause health problems by increasing the risk of heat strokes and the probability of gastrointestinal infections. Electricity plays a critical role in avoiding health issues by powering medical devices, maintaining safe indoor temperatures, refrigerating food and medications, and ensuring the availability of clean drinking water. Furthermore, it enables IT communication systems as well as medical record systems.^c

^a D. Rojas et al. 2025. *Energy Justice Through Energy Storage: Supporting Energy Resilience in Disadvantaged Communities*. Current Sustainable/Renewable Energy Report. 12 (17).

^b ADB. *Tuvalu: Increasing Access to Renewable Energy Project - Second Additional Financing*.

^c K. Milliken, J. Lim, and N. Kuruppu. 2025. *Cooling the Heat Crisis with Energy and Health Solutions*. Asian Development Blog. 10 February.

Multiple Benefits of Building Resilience

Enhancing climate resilience of energy systems offers numerous benefits for all key stakeholders, including energy suppliers, consumers and authorities. Recent studies underscore the substantial socioeconomic advantages, indicating strong cost-effectiveness (Table 2). For instance, the International Energy Agency (IEA) reports that the socioeconomic benefits of flood-resistant energy infrastructure in Asia could be 3–8 times higher than the investment costs.²⁰ Similarly, a cost–benefit research conducted by the Economist Impact and UNICEF shows that a \$1 investment in energy resilience of Pakistan could return up to \$3 in social and economic benefits.²¹ The World Bank estimates that investing approximately \$560 million in additional flood protection measures for power systems in Bangladesh could prevent future losses of up to \$1.6 billion.²² These encouraging findings, which primarily focus on the benefits of flood protection measures, suggest that broader efforts to enhance resilience against a wider range of climate hazards would yield benefits far exceeding their costs.

Energy suppliers can achieve significant economic gains through investments in resilience, despite initial up-front costs. Resilient infrastructure reduces long-term operating expenses by preventing damage from extreme weather events, minimizing disruptions, and reducing maintenance and repair expenditures. Beyond financial savings, resilience measures also enhance supply chain stability, enabling more effective long-term planning and investment decisions.

Consumers benefit directly from resilient energy systems through improved energy access, stabilized energy prices, and an enabling environment for economic growth and social welfare. Reliable energy is critical for expanding universal access to modern services, particularly in Asia and the Pacific where approximately 75 million people still lack electricity. Currently, many of the countries that have the most limited access to reliable electricity in Asia and Pacific are small island developing states—such as Papua New Guinea, Vanuatu, Solomon Islands, and Kiribati—or least developed countries, including Myanmar, Afghanistan, Nepal, and Cambodia.²³ As these countries not only lack access to reliable electricity but are also highly vulnerable to climate impacts, incorporating affordable and sustainable resilience measures into the development of new energy systems is essential. Without climate resilience, newly installed power generation assets and electricity grids may be damaged or disrupted by extreme weather events, preventing the achievement of universal electricity access.

Furthermore, resilient electricity systems contribute to stabilizing electricity prices. In many Asian developing countries, the combination of volatile fuel costs and supply interruptions caused by extreme weather events leads to sharp fluctuations in electricity prices, undermining affordability and economic stability. Climate-resilient electricity systems help buffer these price shocks by reducing the need for fossil fuel imports, which are often used for backup generation, thereby improving affordability and mitigating exposure to global fuel price volatility.

A resilient energy system is a catalyst for economic growth of energy consumers. Reliable electricity underpins industrial activity and commercial productivity. Strengthening resilience not only secures energy supply but also unlocks economic opportunities. In some countries, investing in domestically available energy resources can improve trade balances by reducing reliance on imported fuels, while contributing job creation in local energy industries.

²⁰ International Energy Agency (IEA). 2022. *Climate Resilience for Energy Security*.

²¹ Economist Impact. 2024. *Powering Progress: Measuring the Socioeconomic Costs and Benefits of Investing in Energy Resilience for Healthcare, Education and Water*.

²² World Bank Group. 2019. *Stronger Power: Improving Power Sector Resilience to Natural Hazards*.

²³ IEA et al. 2024. *Tracking SDG7: The Energy Progress Report 2024*.

Moreover, resilient energy systems protect essential social services—such as health care, education, and water supply—from climate-driven disruptions, safeguarding communities. For instance, resilient energy supply plays a central role in protecting people against heat stroke and gastrointestinal infections during extreme heat events, regulating indoor temperature, maintaining cold chains for food and medicine, and ensuring that hospitals and emergency shelters remain operational. In case of drought, resilient energy systems can mitigate water shortage by enabling alternative water supply solutions such as water pumping and desalination.

National authorities will also benefit from climate-resilient energy systems by achieving their objectives related to energy security, accelerating energy transitions, and enhancing disaster risk management.

Resilient energy systems are essential for ensuring stable and reliable energy supply in increasingly variable climate conditions. By incorporating resilience considerations into energy policies, investment planning, and regulations, countries can encourage infrastructure upgrades, enhance energy efficiency, and diversification of energy sources. Enhanced designs and physical resilience of energy infrastructure, such as electricity networks, enable continued energy supply against extreme weather events, withstand increasing temperatures and sea levels, and rapidly restore function after climate-driven disruptions.²⁴ Additionally, more diversified, efficient, and upgraded energy systems can reduce the energy sector’s dependence on imported fuels and minimize exposure to global supply chain disruptions (e.g., delayed oil shipments).

Furthermore, climate resilience supports the acceleration of energy transitions to renewable energy sources. Unlike conventional power sources, renewables are highly sensitive to climatic conditions. Hydropower depends on steady rainfall, solar output fluctuates with temperature and irradiance, and wind farms can be disrupted by extreme heat or cyclones. Without resilience-informed planning, renewable energy systems may underperform or fail. Integrating climate projections into energy planning and investment decisions ensures that transitions to low-carbon energy are sustainable, economically feasible and resilient in the long term.

Resilient energy systems can also play a significant role in disaster preparedness, response, and recovery. Resilient energy systems that are often equipped with advanced digital technologies can detect and prevent disasters, and enable real-time monitoring, early warning systems, and rapid failure detection. It could lower maintenance and repair costs and support fast disaster recovery and emergency communication channels. Furthermore, consistent energy supply supports the implementation of post-disaster actions, such as transportation of food and medicine.

Table 2: Multiple Benefits of Climate-Resilient Energy System

STAKEHOLDERS	 National authorities	 Energy suppliers	 Energy consumers
	KEY BENEFITS		
Multiple Benefits of Climate-Resilient Energy System 	• Strengthened energy security and reduced reliance on imported fuels • Accelerated energy transition • Enhanced disaster risk management and rapid recovery	• Reduced lifetime costs for maintenance and repair • Increased stability and reliability of supply chains	• Reliable energy access • Stable electricity prices • Improved economic growth, productivity, and job creation • Protection of essential social services

Source: Author.

²⁴ Footnote 20.

4 What Are the Key Barriers to Energy Sector Resilience?

As climate impacts intensify, the energy sector in developing countries faces growing challenges, yet efforts to strengthen resilience remain hindered by some critical barriers. One of the most significant barriers is the limited availability of accessible and accurate information on how future climate would affect energy systems. Although climate science and projections have advanced considerably over the past 3 decades, their application within the energy sector remains underdeveloped. Another barrier is the weak integration of adaptation and resilience into national climate and energy strategies. In many cases, energy policies are heavily weighted toward mitigation, often overlooking the need for adaptation. It may hinder national-level planning for a resilient energy system with diversified energy sources and adaptation technologies. Investment in resilient energy systems also falls short, with adaptation finance representing only a small fraction of overall funding. Furthermore, the adoption of innovative technologies that could enhance energy system resilience is progressing slowly in many developing countries, largely due to comparatively high up-front costs despite their long-term benefits. This section examines these critical barriers to building a more resilient energy sector.

Limited Availability of Accessible and Accurate Climate and Weather Information for Energy

The growing climate and weather impacts on energy systems combined with the rapid expansion of renewable energy underscore the urgent need for reliable information to guide decisions in the energy sector. While increasing climate impacts on traditional energy supply and demand highlight the importance of risk consideration in the energy sector, further penetration of climate-sensitive renewable energy sources mandates climate-resilient decision-making. Climate risk and impact assessments based on the latest information become essential, helping project developers and investors identify suitable locations and technologies, minimize risks, and avoid costly disruptions. Climate information is also critical for developing operation and maintenance plans, establishing early warning systems, and designing effective disaster recovery strategies. For example, ADB's Digitize to Decarbonize—Power Transmission Grid Enhancement Project in Uzbekistan enhanced reliability and stability of the power transmission grid by conducting site-by-site climate risk assessment based on robust data. The project integrated adaptation measures based on the assessment results, including more effective cooling systems, resilient conducting materials, improved cables, retention walls near substations, strengthened drainage and slope, and additional transmission circuits for re-routing.²⁵

However, developers and investors frequently report challenges in accessing granular, business-relevant climate data. According to the *State of Climate Change Report* by the Association of Southeast Asian Nations, the inadequate quality of climate observation data and projections in Southeast Asia is a major bottleneck for conducting reliable climate risk and impact assessments at sector level.²⁶ Despite recent progress in access

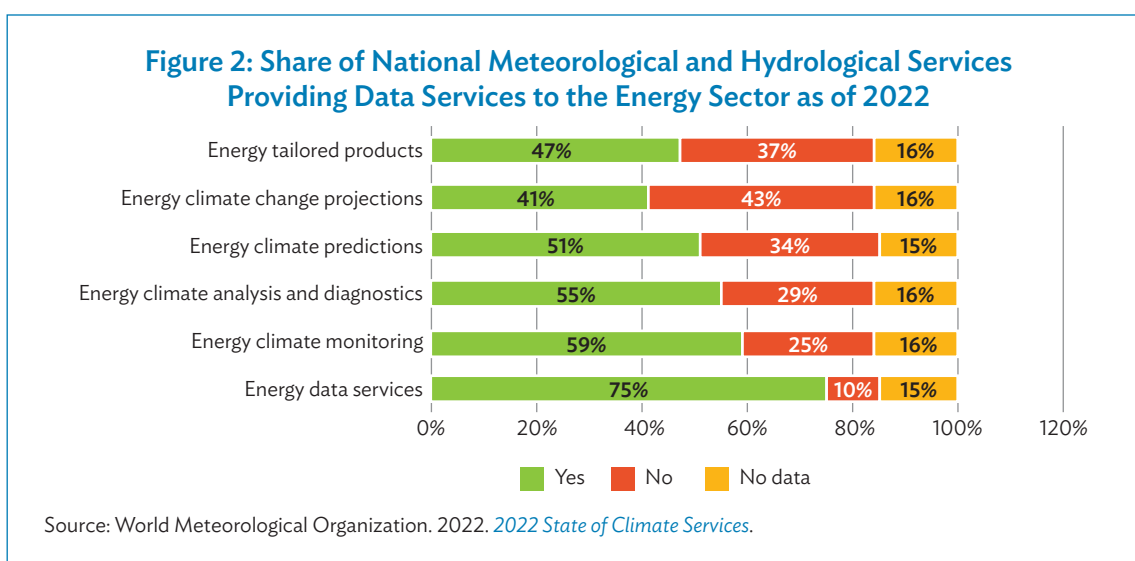
²⁵ ADB. *Uzbekistan: Digitize to Decarbonize—Power Transmission Grid Enhancement Project*.

²⁶ Association of Southeast Asian Nations. 2021. *ASEAN State of Climate Change Report*.

to high-resolution datasets and advanced climate projection models, energy providers are still experiencing difficulties in obtaining quality data. Even when energy providers have access to detailed, forward-looking climate projections, it remains difficult to translate complex probabilistic data into actionable insights for specific energy technologies. For instance, even when a hydropower developer has access to future precipitation data, it requires careful translation of the data to calculate future water availability for hydropower generation. It may require projections of various climate factors such as evapotranspiration and glacier melting in addition to a full set of precipitation data. Due to the challenges in translation of climate data, many developers continue to rely on simple historical observations and low-resolution datasets, which are often inadequate for forward-looking and project-specific analysis.

In addition, government-supported climate services remain limited, despite their critical role in enabling resilient energy planning. Governments can play a pivotal role by supporting the collection and dissemination of scientific data and projections tailored to the energy sector. Collaboration between relevant ministries and meteorological and hydrological services can ensure that climate information is publicly available, appropriately downscaled, and translated into variables relevant to energy sector stakeholders. This would enable more credible and actionable future planning. However, such efforts are currently insufficient. According to a 2022 World Meteorological Organization (WMO) survey, fewer than 50% of member countries provide tailored climate services for the energy sector and only about 40% offer future projections (Figure 2). Services specifically designed for renewable energy are even scarcer, with fewer than 30% of countries maintaining dedicated observation networks for energy applications.²⁷

These findings highlight the urgent need to expand and strengthen weather projection services for the energy sector. Installing weather stations near energy assets, increasing detection measures for solar radiation and wind patterns, and deploying smart technologies and remote-sensing tools can contribute to the collection of information. Some efforts to enhance services for energy systems are already underway. For instance, the Manila Observatory, a nonprofit organization, is providing an automated weather forecast system to support solar and wind energy generation with 5-day forecasts at 5-kilometer spatial resolution.²⁸



²⁷ World Meteorological Organization (WMO). 2022. *2022 State of Climate Services*.

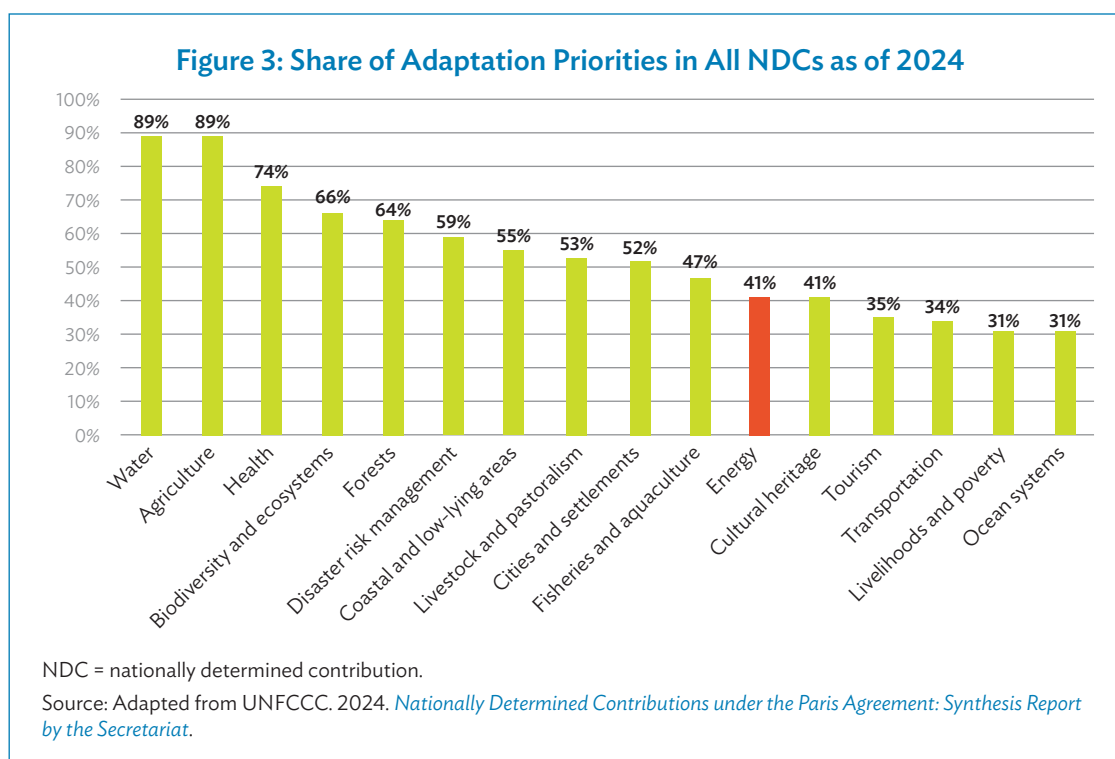
²⁸ Manila Observatory. 2022. *High-Definition Clean Energy, Climate, and Weather Forecasts for Philippines*.

Limited Consideration of Energy System's Adaptation in National Climate Strategies

Adaptation and resilience for energy systems needs to be further integrated into national strategies.

Mainstreaming climate resilience in energy systems through national energy and climate strategies is essential for facilitating adaptation efforts and addressing investment barriers. High-level policy documents developed through consultations with different stakeholders can help initiate and activate discussions on climate impacts and resilience measures among stakeholders. This, in turn, encourages energy sector stakeholders to consider relevant risks from the earliest stages of project development and incorporate resilience measures into operation and maintenance.

However, discussions on energy and climate have traditionally focused on mitigation while adaptation has received significantly less attention. This imbalance is particularly evident in national climate strategies such as nationally determined contributions (NDCs). Mandated by the Paris Agreement, NDCs require countries to prepare, communicate, and maintain their contributions to achieve the Paris Agreement goals. Given their broad coverage and participation, NDCs play a central role in shaping national climate policies and streamlining efforts to reduce emissions and adapt to climate impacts. In these NDCs, however, discussions on adaptation and resilience in the energy sector are still limited (Figure 3). According to the United Nations Framework Convention on Climate Change (UNFCCC), only 41% of NDCs submitted by 2024 prioritized adaptation in the energy sector²⁹ while 95% included mitigation targets for energy.³⁰

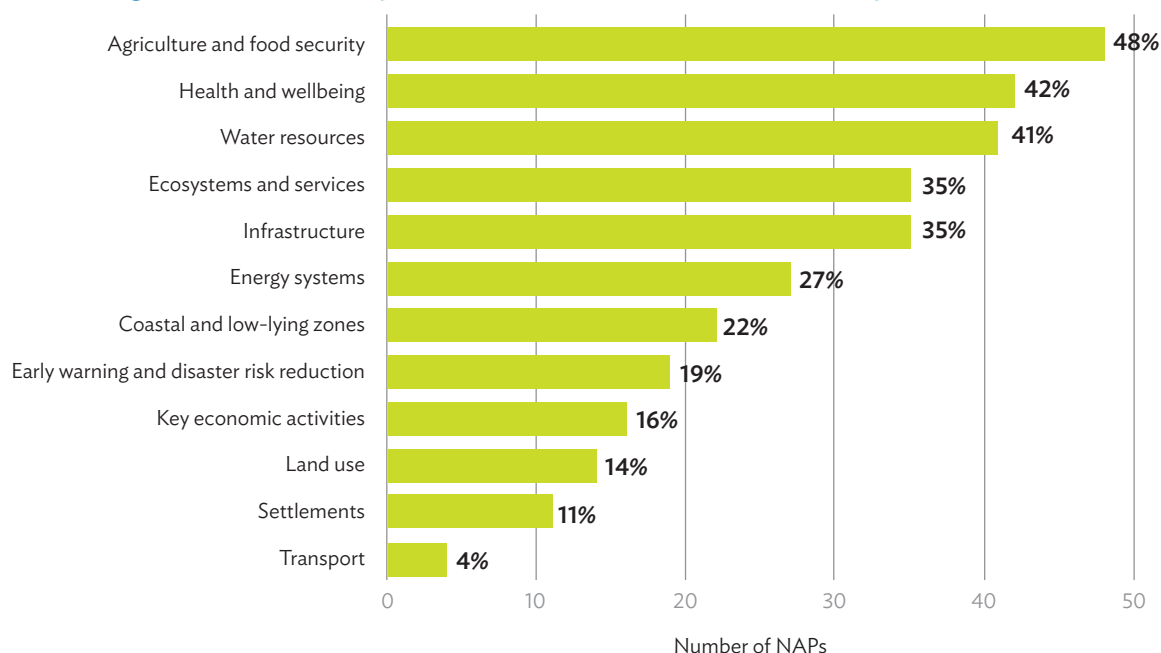


²⁹ UNFCCC, 2024. *Nationally Determined Contributions under the Paris Agreement Synthesis Report by the Secretariat*; and WMO, 2022. *2022 State of Climate Services*.

³⁰ UNFCCC, 2024. *Nationally Determined Contributions under the Paris Agreement: Synthesis Report by the Secretariat*.

The neglect of the energy sector’s adaptation is also evident in national adaptation plans (NAPs). As part of the UNFCCC process, NAPs aim to reduce vulnerability to climate impacts by building adaptive capacity and resilience and facilitating the integration of climate considerations into relevant new and existing policies, programs and activities, in particular development planning processes and strategies.³¹ NAPs lay out the actions that a country plans to follow, and typically include a description of climate risks, adaptation priorities and actions, existing adaptation barriers, and identified adaptation measures. While agriculture, health, water, and ecosystem are underscored in NAPs, energy has been often overlooked. As of 2023, only 27% of global NAPs (13 out of 48) recognized energy as an adaptation priority³² despite the growing importance of the energy sector’s resilience in many countries (Figure 4).

Figure 4: Share of Adaptation Priorities in All NAPs as of 30 September 2023



NAP = national adaptation plan.

Source: UNFCCC. 2023. *National Adaptation Plans 2023: Progress in the Formulation and Implementation of NAPs*.

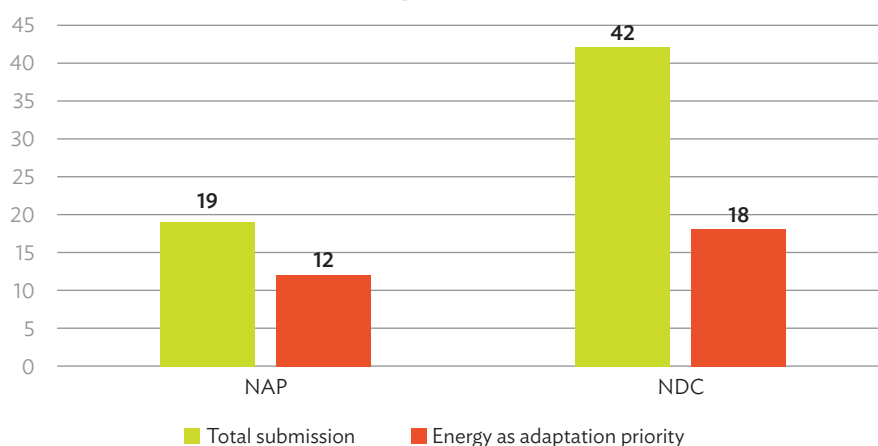
As climate impacts intensify, efforts to improve adaptation and resilience of energy systems are increasing. Some countries have begun incorporating energy sector adaptation into their national strategies and action plans. For example, the European Union acknowledged the increasing climate risks to all components of the energy system and proposed actions such as improving water management for hydropower and thermal power plants, integrating adaptation strategies into monitoring procedures, and enhancing water efficiency in energy production.

³¹ UNFCCC. *National Adaptation Plans*.

³² UNFCCC. 2023. *National Adaptation Plans 2023: Progress in the Formulation and Implementation of NAPs*.

The Asia and Pacific region has made progress but requires further efforts. As of 20 October 2025, 18 out of 42 ADB developing member countries (DMCs) prioritized adaptation in the energy sector—or broader infrastructure, including energy—in their NDCs. Additionally, 12 out of 19 DMCs that have submitted NAPs to the UNFCCC identified energy as an adaptation priority (Figure 5). Although these efforts align with, or even exceed, global trends, further adaptation discussions are required in Asia and the Pacific to address the rapidly increasing climate impacts in the region. The combined effects of soaring energy demand, aging infrastructure, and climate impacts necessitate further adaptation measures in the energy sector. Strengthening the energy sector’s representation in national adaptation priorities is key to reducing the energy sector’s vulnerability to climate impacts.

Figure 5: Share of NDCs and NAPs Including Energy as Adaptation Priority, ADB DMCs



ADB = Asian Development Bank, DMC = developing member country, NAP = national adaptation plan, NDC = nationally determined contribution.

Note: The values are calculated by the author based on NDCs and NAPs submitted to the United Nations Framework Convention on Climate Change as of 20 October 2025.

Source: UNFCCC. 2023. *National Adaptation Plans 2023: Progress in the Formulation and Implementation of NAPs*.

Insufficient Adaptation Finance in the Energy Sector

Current adaptation finance falls significantly short of investment needs. Approximately \$34 billion of tracked adaptation finance from both public and private sectors was committed in Asia and the Pacific in 2021–2022.³³ Most of this amount comprised domestic expenditure in the People’s Republic of China (PRC), while finance flows for international adaptation in the region were approximately \$11 billion.³⁴ The current adaptation finance flow is way less than adaptation needs, which are estimated to be between \$102 billion (based on sector modeling) and \$431 billion (based on stated adaptation needs in NDCs, NAPs, adaptation communications, and NCCCs) per year by 2030.³⁵

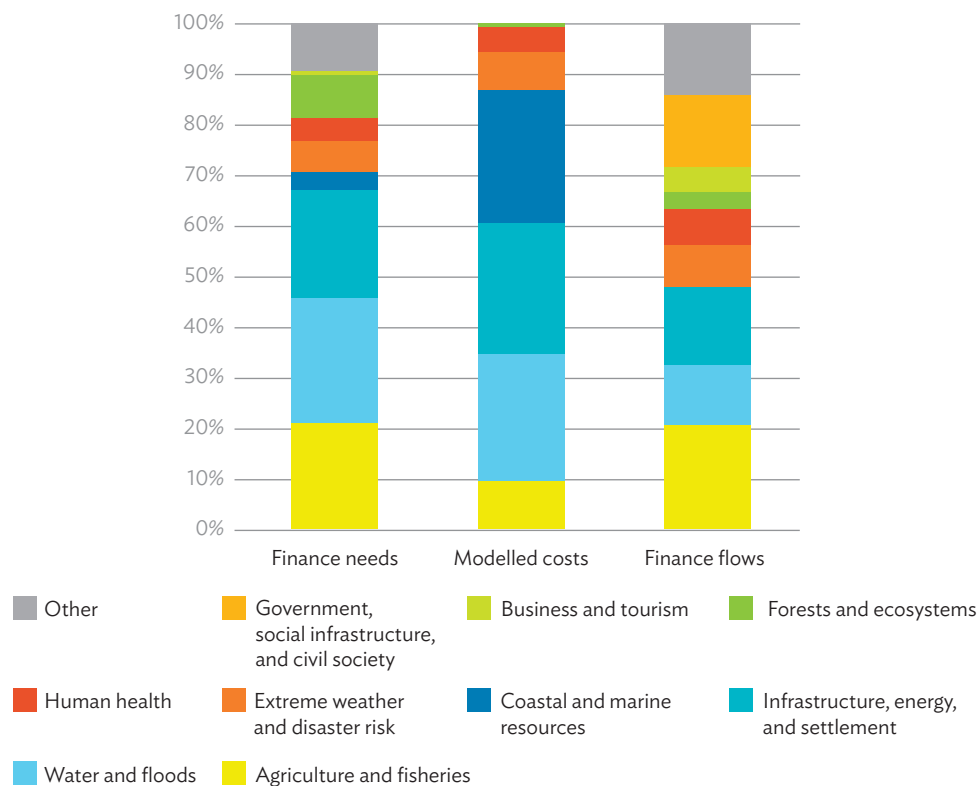
³³ Footnote 19.

³⁴ Climate Policy Initiative and Global Center on Adaptation. 2024. *State and Trends in Climate Adaptation Finance 2024*.

³⁵ Footnote 19.

Adaptation finance flow in the energy sector remains insufficient as well. Building resilient energy infrastructure is one of the areas which requires the largest amount of adaptation financing. The *UNEP Adaptation Gap Report 2024* estimates more than \$56 billion per year will be needed for resilient energy and transport infrastructure, and the adaptation needs may increase significantly toward 2050.³⁶ It affirms that “infrastructure, energy and settlement” is one of the sectors which requires the largest finance needs.³⁷ However, adaptation finance flows for the energy sector remains short, as most climate finance goes to mitigation. For instance, in 2023, multilateral development banks (MDBs) committed only \$7.25 billion for adaptation in energy, transport, and other built environment and infrastructure, which accounts for less than 10% of total climate finance committed to the sector (\$76.45 billion).³⁸ Although low- and middle-income countries tend to invest more in adaptation of infrastructure (18%) than high-income countries (2%), it is still insufficient. The report shows that the share of “infrastructure, energy and settlement” in international public adaptation finance flow for developing countries remains small (Figure 6) compared to financial needs.³⁹

Figure 6: Comparison of Adaptation Finance Needs (extrapolated) and Modelled Costs of Adaptation (annual to 2030) and International Public Adaptation Finance Flow (2018–2022 average) for Developing Countries by Sector



Source: United Nations Environment Programme. 2024. *Adaptation Gap Report: Come Hell and High Water*. <https://www.unep.org/resources/adaptation-gap-report-2024>.

³⁶ Global Center on Adaptation and Climate Policy Initiative. 2023. *State and Trends in Climate Adaptation Finance 2023*; UN Environment Programme. 2023. *Adaptation Gap Report 2023*.
³⁷ United Nations Environment Programme. 2024. *Adaptation Gap Report: Come Hell and High Water*.
³⁸ European Investment Bank (EIB), et al. 2024. *2023 Joint Report on Multilateral Development Banks' Climate Finance*.
³⁹ Footnote 38.

Private capital has been significantly limited in adaptation finance due to barriers related to market failures.

At the global level, only 1.6% of adaptation investment (around \$500 million) came from the private sector in 2017–2018.⁴⁰ In Asia and the Pacific, the private sector's share is even smaller, at just 0.2% (approximately \$84 million) in 2018–2019⁴¹ and around 1% in 2022–2023.⁴²

Moreover, the majority of private adaptation finance is concentrated in water and sanitation, with negligible flows directed toward adaptation and resilience in the energy sector.⁴³ This negligence of energy sector adaptation in private finance poses a significant concern, as public finance alone cannot meet the energy sector's urgent resilience needs.

While enhancing resilience in energy systems benefits society as a whole, the costs are unevenly distributed across the energy value chain, raising the question of who should bear them. Although private energy providers have a direct interest in protecting their own assets and ensuring reliable service delivery, they may perceive the up-front costs of resilience as outweighing the returns, particularly when the indirect benefits of reliable energy supply are not captured in their investment decisions. Private actors can also be discouraged when the benefits of resilience only become tangible over the long term, making it difficult to justify immediate investment, especially when future returns are uncertain.⁴⁴

These challenges are even more pronounced in developing countries. Private investors often face additional barriers such as the lack of reliable climate risk data, which complicates efforts to assess and price risks accurately. Monopolistic market structures in many developing countries further reduce incentives for investment in resilience. Policy uncertainties and misalignments and weak enabling financial markets are other constraints to raising private capital for adaptation.⁴⁵

Slow Progress in Adopting Innovative Technologies

Despite substantial progress in resilience technologies, the pace of their deployment in developing countries remains slow. Adoption of innovative technologies is heavily concentrated among a few countries. Two-thirds of all high-value patented climate adaptation technologies in 2010–2015 were concentrated in the PRC, Germany, Japan, the Republic of Korea or the United States.⁴⁶ While the geographic concentration of innovation does not necessarily prevent deployment elsewhere, many developing countries face structural barriers that hinder adoption. These include low awareness, limited technical capacity, high financing costs, weak institutional frameworks, and the need to adapt technologies to local contexts.⁴⁷

In the energy sector, the uptake of innovative adaptation technologies has lagged behind, despite many of these technologies being both mature and economically viable. For instance, variable renewable energy technologies that can support climate resilience by enabling diversification and decentralization of energy sources remain limited in many developing countries. While the global share of variable renewables in electricity generation

⁴⁰ World Bank Group. 2021. *Enabling Private Investment in Climate Adaptation and Resilience: Current Status Barriers to Investment and Blueprint for Action*.

⁴¹ Footnote 19.

⁴² Footnote 19.

⁴³ Climate Policy Initiative. 2024. *Tracking and Mobilizing Private Sector Climate Adaptation Finance 2024*.

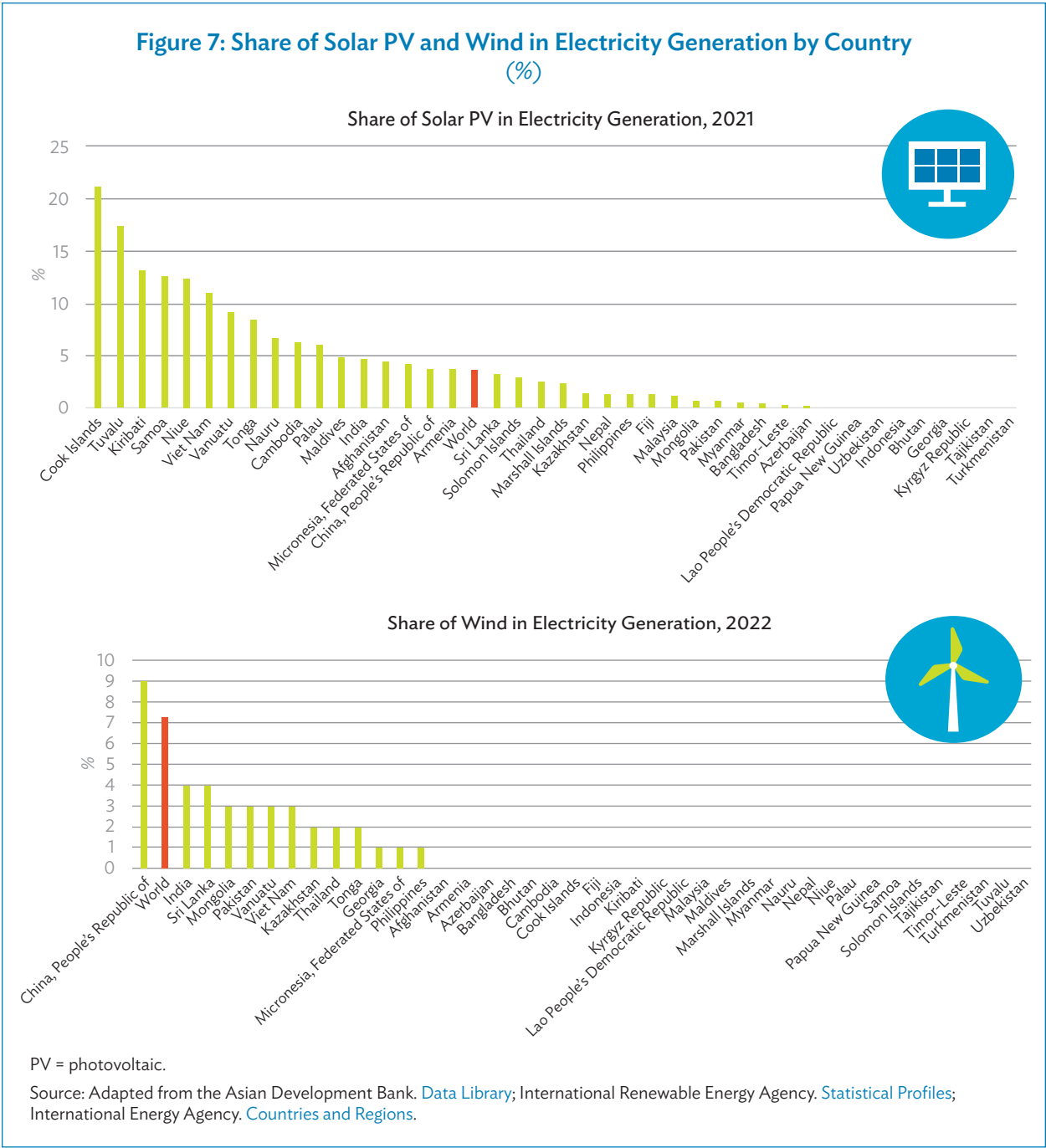
⁴⁴ UNEP. 2023. *State and Trends in Climate Adaptation Finance 2023*.

⁴⁵ Footnote 19.

⁴⁶ A. Dechezleprêtre, et al. 2020. *Invention and Global Diffusion of Technologies for Climate Change Adaptation: A Patent Analysis*. World Bank Group.

⁴⁷ United Nations Environment Programme Copenhagen Climate Centre. 2023. *Technology Transfer for Climate Mitigation and Adaptation*.

increased notably from 1.7% in 2010 to 9.1% in 2020 driven largely by falling costs, most developing countries in Asia and the Pacific still fall below the global average in solar photovoltaic (PV) system and wind power penetration rates (Figure 7). In 2021, 24 out of 41 ADB DMCs had a lower share of solar PV in their electricity mix than the global average of 3.6%. In terms of wind power, all ADB DMCs except the PRC fell below the 2022 global average of 7.3%.⁴⁸



⁴⁸ ADB. 2021. *Key Indicators for Asia and the Pacific 2021*; International Renewable Energy Agency (IRENA). 2022. *Renewable Energy Statistics*; International Energy Agency (IEA). 2024. *Renewables 2024*.

Smart grids offer another example of the areas that require further penetration of innovative resilience technologies. Smart grids are electricity networks equipped with digital and other advanced information and communication technologies to monitor and manage the transport of electricity from all generation sources to meet the varying electricity demands of end users. They are crucial for building resilient electricity systems, as they enhance real-time visibility, allow operators to respond quickly to disruptions, and support faster power restoration. When disruptions occur, smart grids can automatically alert the operators and enable them to reroute power around the detected problem, preventing lengthy outage and facilitating restoration. Furthermore, remote sensing and monitoring technologies can prevent climate-driven disruptions by enabling early warning and emergency actions in advance. However, the deployment rates of smart grids vary widely across countries. For instance, the PRC had achieved 100% of residential smart meters by 2021 when India was at 2%.⁴⁹

Deployment of innovative and efficient solutions for space cooling also remains low. Energy-efficient cooling solutions can enhance the resilience of electricity systems, especially as rising temperatures and more frequent heatwaves drive up electricity demand. The augmenting cooling demand can add strains to energy systems in developing countries in Asia and the Pacific, which are already stretched out due to rapid demand increase with population and economic growth. Solutions such as highly efficient air conditioners, passive cooling designs, cool roofs, and district cooling systems can significantly reduce energy consumption and ease pressure on already strained power systems. However, in many cases, energy consumers in developing countries tend to opt for less efficient cooling options even when higher-efficiency solutions are available at comparable prices. For instance, according to the IEA's research, Southeast Asian customers purchase air conditioners that are significantly less efficient than what is available while the best available air conditioning technologies are already more than twice as efficient as the market average in Southeast Asia.⁵⁰

⁴⁹ IEA. 2023. *Deployment to Date of Residential Smart Meters, 2021*.

⁵⁰ IEA. 2019. *The Future of Cooling in Southeast Asia*.

5 How Can the Energy Sector Overcome Key Barriers and Build Resilience?

Efforts to enhance climate resilience in the energy sector remain constrained by the lack of quality climate information for energy sector, limited consideration of energy systems in national adaptation priorities, inadequate adaptation finance, and slow uptake of innovative technologies. Developers and investors often struggle to access granular, business-relevant climate data, highlighting the urgent need for improved climate information. Although national climate strategies increasingly emphasize the importance of adaptation, they often fall short of addressing the specific needs of the energy sector. As a result, adaptation finance for the energy sector remains critically insufficient, and the deployment of resilience technologies in developing countries continues to lag.

To overcome these barriers and strengthen resilience, actions from all key stakeholders are essential. Energy and climate authorities play a pivotal role in improving the accessibility and quality of information and early mainstreaming of energy sector resilience into regulations and policies. Energy suppliers are expected to lead investments in resilient infrastructure and accelerate the deployment of adaptation technologies. Energy consumers can support these actions through voluntary contributions to climate data collection, political support, behavioral changes, and demand-side solutions. With collaborative efforts to enhance the climate resilience of energy systems, stakeholders can achieve a wide range of adaptation benefits, including improved reliability and accessibility of energy services, economic growth through increased productivity and job creation, enhanced social welfare, stronger energy security, a faster energy transition, and more effective disaster risk management.

Improved Quality of Climate Information for Energy Systems

Climate information is crucial for building resilience. With accurate and accessible data, project developers, investors, and operators can conduct climate risk assessments and optimize energy system design and operation while consumers can adapt their energy use accordingly. However, in many developing countries in Asia and the Pacific, climate information for the energy sector remains limited. Energy project developers and investors frequently report challenges in accessing and obtaining quality information despite notable progress in downscaling and accuracy in climate projections in recent years. Translating complex probabilistic data into actionable insights for specific energy technologies still requires high technical capacity and resources while public climate services are failing to fill this gap.

Government authorities, including relevant ministries and agencies, can address these knowledge barriers by enhancing the quality of climate information services. While most countries have national meteorological and hydrological agencies that serve as the primary sources of climate data, the information they provide often falls short in terms of resolution, accuracy, projections, coverage, and level of detail. As a result, it may not meet the requirements for effective climate risk and impact assessments that could guide site selection, or the design,

operation, and maintenance of infrastructure. Recognizing this challenge, some governments have already launched initiatives to provide high-quality climate information using the latest techniques in climate modeling, which can be used along with sector-level information on exposure and vulnerability to guide climate risk assessment. For instance, Centre for Climate Research Singapore produced detailed climate projections with a spatial resolution of 8 kilometers for all of Southeast Asia using IPCC data and dynamic downscaling.⁵¹ Malaysia developed the Malaysian Adaptation Index, which summarizes the level of climate risks and readiness of each Malaysian state and is supported by localized analyses of future projections.⁵²

Government authorities can close knowledge gaps by strengthening climate observation networks. By installing observation stations near major energy assets—such as power plants, substations, refineries, and energy storage facilities—governments can lay the knowledge foundation for more customized climate services for the energy sector and gain a better understanding of the specific climate hazards facing each asset. Government’s investment in advanced technologies such as remote sensing, satellite analysis, and real-time monitoring can also enable system-wide understanding and support more comprehensive adaptation planning. In this context, the World Meteorological Organization, UNDP, and UNEP have established the Systematic Observations Financing Facility to bridge the gap by providing grant funding for the collection of weather and climate data in countries with the most severe shortfalls in observations.⁵³

Energy suppliers can contribute to improving the quality of climate information by conducting robust climate risk and impact assessments. Utilities, grid operators, project developers, and private investors can contribute to sector-specific data collection through conducting detailed on-site climate risk assessments and technical analyses. The findings create a virtuous cycle by feeding assessment results back into national climate information systems. For instance, Perusahaan Listrik Negara, Indonesia’s state-owned electric utility, developed a risk assessment portal that provides information on the areas of climate hazards and affected populations’ potential physical and economic losses, by overlaying government’s risk assessment data from the National Agency for Disaster Countermeasures with asset condition data.⁵⁴ It helps the state electricity company integrate climate resilience consideration in planning and enhance its response to climate-related disasters. This example shows how energy suppliers can contribute to advancing climate information while utilizing data provided by national authorities.

Energy consumers can improve the quality of information by providing data on climate impacts from the demand side. Consumers can offer valuable feedback on how climate and weather influence their energy consumption patterns and comfort levels. Such information can be used to refine setpoints for energy consumption and sector-specific indicators for monitoring climate impacts on the energy sector (e.g., thresholds for cooling degree days and heating degree days). Furthermore, businesses and households may install energy consumption monitoring systems that track real-time energy usage alongside temperature data. This can enable a more detailed analysis of the relationship between climate variables and energy demand patterns. For instance, Australian Energy Market Operator advanced its electricity demand forecasting using the real-time energy consumption data. The company collects household electricity consumption data every 30 minutes using smart metering technologies and compares the data with temperature data from historical observations and climate projections. It helps them set up robust thresholds for cooling and heating demands and improve accuracy of electricity demand forecasting.⁵⁵ Such data helps the energy sector better understand climate-induced shifts in consumption patterns and supports system-wide assessments of climate impacts on energy use (Table 3).

⁵¹ Centre for Climate Research Singapore. 2024. *Singapore’s Third National Climate Change Study*.

⁵² Institut Penyelidikan Air Kebangsaan Malaysia. *Malaysia Adaptation Index*.

⁵³ The Systematic Observations Financing Facility. <https://un-soff.org/about/>.

⁵⁴ Esri. 2025. *PLN Innovates Network Vulnerability Assessment with GIS*.

⁵⁵ Australian Energy Market Operator. 2025. *Electricity Demand Forecasting Methodology*.

Table 3: Actions for Improved Quality of Climate Information for Energy Systems by Stakeholder

STAKEHOLDERS			
	National authorities	Energy suppliers	Energy consumers
ACTIONS			
Actions for Improved Quality of Climate Information for Energy Systems by Stakeholder 	- Enhance the quality of public climate information services at national, regional, and local levels - Strengthen climate observation networks by expanding observation stations near major energy assets and introducing advanced technologies	- Conduct robust climate risk and impact assessments - Feed assessment results back into national climate information systems	- Offer valuable feedback on how climate and weather influence their energy consumption patterns and comfort levels - Install energy consumption monitoring systems to enable more detailed analysis of climate variables and energy demand patterns

Source: Author.

Incorporation of Climate-Resilient Energy Systems in National Strategies

To foster climate-resilient energy systems, national policies have a critical role to play. Despite clear incentives for energy suppliers to safeguard energy infrastructure against climate-driven disruptions, they may underinvest in resilience, as they shoulder only a small share of the total climate-related losses while society bears most of the costs. This underinvestment can be compounded by perceived high capital costs, long return periods, and monopolized market conditions. Effective policy measures are, therefore, needed to avoid such market failures and enforce resilience measures that protect broader socioeconomic benefits.

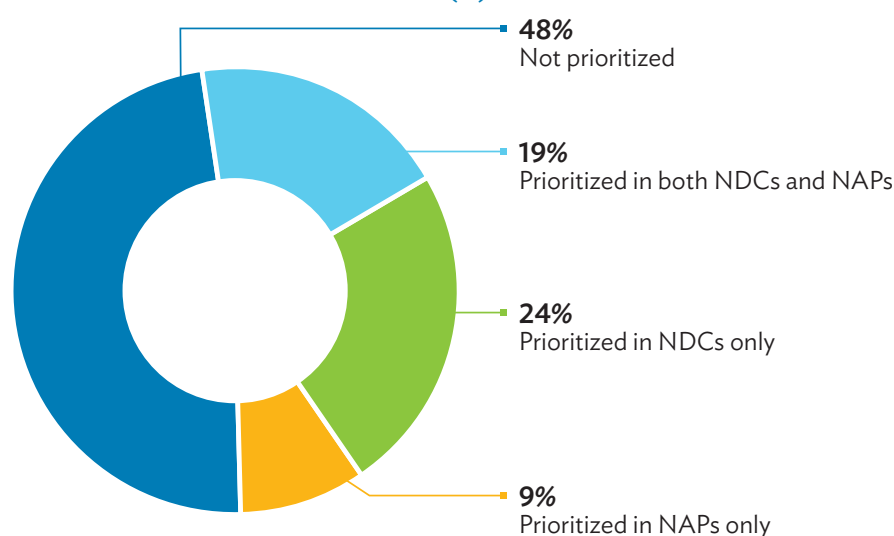
Recognizing the energy sector as an adaptation priority in national climate strategies is the first step.

Addressing energy sector climate resilience in high-level policy documents can raise awareness of and activate discussions on climate risks and resilience measures among stakeholders. It can encourage energy suppliers to consider climate risks and impacts from the early stages of projects and incorporate resilience measures throughout the project lifecycle. It can also improve understanding of climate risks among energy consumers, prompting voluntary actions to support resilience measures in the energy sector.

Countries have already made efforts to incorporate climate resilience in their national climate strategies, although these efforts remain insufficient.

As of 20 October 2025, 22 out of 42 ADB DMCs included the energy sector in their NDCs or NAPs as an adaptation priority. In some countries such as Armenia, Cambodia, Fiji, Kiribati, the Lao PDR, the Marshall Islands, Nepal, and the Philippines, both NDCs and NAPs prioritize the energy sector for adaptation actions while energy sector adaptation in other 14 countries is addressed in either the NDCs or the NAPs. However, nearly half of ADB’s DMCs have yet to integrate the energy sector into their adaptation strategies even though most have concrete adaptation plans for other sectors or have introduced mitigation measures in the energy sector (Figure 8).

Figure 8: Share of ADB DMCs That Prioritized Energy as an Adaptation Priority in NDCs and NAPs (%)



ADB = Asian Development Bank, DMC = developing member country, NAP = national adaptation plan, NDC = nationally determined contribution.

Note: The values are calculated by the author based on NDCs and NAPs submitted to the United Nations Framework Convention on Climate Change as of 20 October 2025.

Source: Author.

National authorities, including governments and regulators, are expected to lead efforts to reflect the importance of climate resilience in the energy sector within national climate strategies. Energy and climate authorities should organize consultations with a wide range of stakeholders to identify concrete actions to enhance climate resilience in the sector. Since prioritization alone does not guarantee strengthened resilience, resilience measures across the entire lifecycle of energy projects should be formalized as an implementation plan. An effective implementation plan may include a list of adaptation measures, a clear timeline for implementation, well-defined stakeholder responsibilities, a strategy for funding and budget allocation, performance indicators, and a robust monitoring and evaluation framework.

Cambodia's Third Nationally Determined Contribution provides a strong example, outlining specific adaptation measures to enhance energy sector resilience. It identifies specific adaptation measures for the three priority areas of energy sector resilience: enhancing climate-proof energy infrastructure, strengthening grid resilience and monitoring, and enhancing preparedness and capacity building. The plan also designates responsible ministries and supporting agencies, while establishing specific performance targets of reinforcing 100% of high-risk assets by 2030 and halving climate-related outages, strengthening 80% of high-priority infrastructure, improving grid resilience by 2030, and achieving less than 30-minute response time for 90% of warnings by 2035.⁵⁶

⁵⁶ Kingdom of Cambodia. 2025. *Cambodia's Third Nationally Determined Contribution (NDC 3.0)*.

NAPs that are aligned with NDCs can serve as an implementation plan for adaptation measures identified in NDCs. According to the UNFCCC, 52% of parties have linked their NAPs and NDCs, with some using NAPs as detailed implementation road maps.⁵⁷ The Lao PDR for instance, recognized energy as a priority sector for climate adaptation in its NDC in 2021 and outlined specific actions for promoting climate resilience through energy security in its NAP in 2025. The plan identified relevant policies, adaptation strategies and objectives for the sector, specific activities, mainly responsible parties, targeted period of implementation, indicators, and targets. By providing a concrete action plan, the Lao PDR's NAP complemented the broader commitments in its NDC. This approach offers valuable insights for other countries in Asia and the Pacific developing or updating their NAPs,⁵⁸ encouraging them to integrate concrete implementation plans for climate adaptation in the NAPs.

International efforts are also being made to support mainstreaming adaptation and resilience. For instance, the Global Goal on Adaptation of the United Nations Framework Convention on Climate Change (UNFCCC) provides a set of indicators to track progress on climate resilience and adaptation which can be applicable for national performance indicators. For the infrastructure sector, the Global Goal on Adaptation proposes “the number of beneficiaries of infrastructure that integrates adaptation measures and/or enables adaptation” and “reduced service disruptions or reduced damage, or increased benefits provided due to climate-resilient infrastructure services that integrate adaptation measures and/or enable adaptation.”⁵⁹

Besides authorities, energy suppliers are expected to play an important role as they are at the center of the real-world implementation of proposed adaptation measures. Investors, utilities, and project developers will be required to embed resilience considerations into project design and operations, in line with the recommendations outlined in national climate strategies. New or revised policies may also call for further actions from energy suppliers, such as aligning their corporate strategies with established national resilience goals. Energy suppliers can contribute to the development and effective implementation of climate resilience policies by providing technical feedback and sharing best practices. Continued dialogues between national authorities and energy suppliers help refine adaptation policies, ensuring they are realistic, implementable, and closely aligned with operational realities.

Energy consumers also play a vital role by supporting and reinforcing government efforts. Communities and civil society can advocate resilience by drawing attention to the socioeconomic costs of climate-induced disruptions that are often overlooked or underrepresented in utility-driven planning. The collective voice of energy consumers can provide a strong political foundation for advancing climate resilience policies and regulations. Consumer engagement also helps to raise public awareness of climate risks and fosters broader acceptance of adaptation measures. In this way, energy consumers complement the actions from energy authorities and suppliers by shaping demand for resilient infrastructure providing public support for adaptation goals (Table 4).

⁵⁷ UNFCCC. 2024. *NDC Synthesis Report*.

⁵⁸ See, for example, UNFCCC. 2023. *Overview - National Adaptation Plans*. Initial guidelines for the preparation of NAPs were developed based on agreements at the seventeenth session of the Conference of the Parties of the Framework Convention on Climate Change, 28 November to 11 December 2011.

⁵⁹ ADB, et al. 2024. *Common Approach to Measuring Climate Results: Update on Indicators*.

Table 4: Actions for Incorporation of Climate-Resilient Energy Systems in National Strategies by Stakeholder

STAKEHOLDERS	National authorities	Energy suppliers	Energy consumers
ACTIONS			
Actions for Incorporation of Climate-resilient Energy Systems in National Strategies by Stakeholder 	• Include the energy sector in national adaptation strategies as one of the adaptation priorities • Organize stakeholder consultations to identify concrete actions across the entire lifecycle of energy projects • Develop a robust implementation plan	• Embed resilience considerations into project design and operations • Align corporate strategies with national adaptation goals • Provide technical feedback and share best practices for further development and effective implementation of adaptation policies	• Provide a strong political foundation for advancing climate resilience policies and regulations • Raise public awareness of climate risks to the energy sector • Foster broader acceptance of adaptation measures

Source: Author.

Scaling Up Adaptation Finance in the Energy Sector

The private sector plays a central role in energy sector investment. According to the International Energy Agency (IEA), the private sector's finance accounted for approximately 75% of total global energy sector investment in 2024.⁶⁰ Private finance has grown particularly in areas directly linked to climate mitigation, such as low-emissions power generation, grid infrastructure, and energy storage, resulting in an increase in the private sector's contribution to climate finance. This trend is also observed in many developing countries where private climate finance has recorded notable growth, largely driven by increased mitigation investments. In Central Asia and South Asia, for instance, private finance represented more than 50% of total mitigation finance between 2018 and 2022.⁶¹ Given that the private sector's share in energy investment is increasing, integrating and scaling up adaptation and resilience measures within its investment decisions is essential to safeguard the long-term reliability of energy services and protect returns on investment.

Private energy suppliers and investors can expand adaptation finance by embedding climate risk assessments and resilience benefits into their decision-making processes. Increasingly, utilities and investors are internalizing climate risks into asset valuation and lifecycle cost analysis, quantifying the benefits of adaptation, and tracking their returns over the entire lifetime of assets. This approach enables them to better understand the financial impact of climate hazards and enhance long-term revenues by reducing climate-related disruptions and physical damage to assets. By considering and monetizing adaptation-related costs and benefits, investors can avoid the market failures of overlooking long-term benefits and justify the up-front costs of resilience measures. For instance, the Energy Development Corporation, the largest energy company in the Philippines, improved reliability of its services and protected future revenues through investing in resilience measures in advance. It embedded climate and geophysical risks into decision-making to cope with the increasing

⁶⁰ International Energy Agency (IEA). 2025. *World Energy Investment 2025*.

⁶¹ Climate Policy Initiative. 2024. *Global Landscape of Climate Finance 2024*.

intensity of tropical cyclones and heavy rainfall.⁶² The Energy Development Corporation spent about \$6.3 million in 2018 on climate adaptation and disaster risk reduction measures based on robust climate risk assessment, and the long-term benefits of climate resilience were estimated to be nearly \$90 million.⁶³

National authorities with public investment on concessional terms play a crucial role, particularly in the early stages of adaptation projects. A blended finance approach—combining concessional capital from multilateral development banks (MDBs), national donor agencies, and philanthropies with commercial investments—can help lower costs, boost investor confidence, and facilitate implementation.⁶⁴ Concessional finance can foster an enabling environment for commercial investment by providing technical assistance for climate risk assessments and demonstrating the viability of new technologies. Additionally, it can attract commercial investors by offering first-loss facilities and guarantees. Such catalytic instruments facilitate the blending of private and public finance by lowering the perceived initial or lifecycle costs for adaptation measures. For example, the Climate Resilience and Adaptation Finance and Technology Transfer Facility integrates a standard growth equity investment fund structure with technical assistance to enable the deployment of climate resilience services and technologies in developing countries. It consists of commercial funding, concessional funding, and a separate grant-based technical assistance facility and created a pipeline of over 500 companies and delivering returns for commercial investors.⁶⁵

Multilateral development banks have contributed to the international community's efforts to fill the adaptation financing gap. The Updated Joint MDB methodology for Tracking Adaptation Finance highlights that adaptation is no longer viewed purely as an add-on to development investments, but rather as an imperative for putting development on the path to resilience.⁶⁶ It underlines that adaptation support and financing procedures for adaptation finance have broadened (Figure 9). As of 2023, MDBs committed to \$24.7 billion for adaptation finance in low- and middle-income economies, accounting for 33% of total climate finance.⁶⁷ This shows a notable increase from \$4.8 billion in 2013, which was equivalent to 20% of total climate finance.⁶⁸ The adaptation finance in infrastructure (including energy, transport, and other built environment) also increased, reaching \$6.36 billion in 2023 which is more than 4 times the level in 2013. In line with the collective efforts of MDBs, ADB has also increased its adaptation finance in the energy sector (Box 3).

⁶² World Bank Group. 2021. *Enabling Private Investment in Climate Adaptation and Resilience: Current Status Barriers to Investment and Blueprint for Action*.

⁶³ Philippine Energy Development Corporation. 2018. *Bouncing Back Strong as Consumers Make the Switch to Renewable Energy*.

⁶⁴ Climate Policy Initiative. 2024. *Building Financial Instruments for Climate Adaptation*.

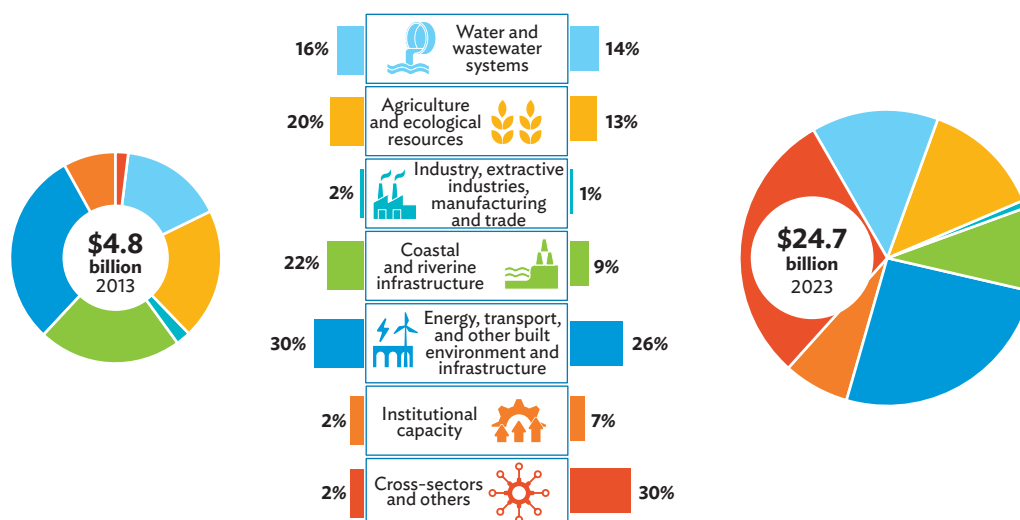
⁶⁵ Climate Finance Lab. Undated. *Climate Resilience and Adaptation Finance & Technology Transfer Facility (CRAFT)*.

⁶⁶ European Investment Bank, et al. 2022. *Joint Methodology for Tracking Climate Change Adaptation Finance*.

⁶⁷ African Development Bank, et al. 2024. *2023 Joint Report on Multilateral Development Banks Climate Finance*.

⁶⁸ African Development Bank et al. 2014. *2013 Joint Report on Multilateral Development Banks Climate Finance*.

Figure 9: MDB Adaptation Finance by Sector in Low- and Middle-Income Economies, 2013 and 2023



Source: Adapted from African Development Bank, et al. 2014. *2013 Joint Report on Multilateral Development Banks Climate Finance*; and African Development Bank, et al. 2024. *2023 Joint Report on Multilateral Development Banks Climate Finance*.

Box 3: ADB's Commitment to Adaptation Finance in the Energy Sector

The Asian Development Bank (ADB) has contributed to global efforts to close the climate finance gap. In 2024, ADB committed for its climate finance to reach 50% of its total annual committed financing by 2030, as it progresses towards delivering over \$100 billion in cumulative climate finance from its own resources from 2019 to 2030. From 2019 to 2025, ADB reached \$55.4 billion toward this commitment, channeling investments across its operations through regular and concessional ordinary capital resources, as well as Asian Development Fund grant resources. The cumulative adaptation finance reached \$21.4 billion overall, lifting ADB's annual adaptation finance from \$1.6 billion (2019) to \$5.3 billion (2025).^a

In the energy sector, the demand for adaptation finance grows as climate impacts intensify. ADB's Climate Change Action Plan, 2023–2030 identified relevant actions for adaptation in the energy sector, including scaling up investment in transformation-enabling infrastructure, retrofitting climate-vulnerable infrastructure, and climate-smart infrastructure design standards and product-agnostic approaches to identifying high-impact climate interventions. ADB's Energy Policy included climate and disaster resilience as one of its principles, specifying its support for developing member countries in building higher resilience with more robust transmission and distribution networks and deployment of various kinds of energy storage to enhance system flexibility.

Reflecting the increasing awareness of climate resilience, energy projects are increasingly transitioning from a narrow focus on climate-proofing toward broader climate resilience outcomes. Adaptation is no longer merely an “add-on” to development investments; neither is it confined to managing physical climate risks to ensure project functionality. Instead, more energy projects are embedding resilience into development pathways, reducing vulnerability, and enhancing adaptive capacity. As a result, the amount of adaptation finance in the energy sector of ADB nearly tripled from \$321.4 million (sum of 2019–2021) to \$925.8 million (sum of 2022–2024). This growth has been driven by the increasing need to integrate climate considerations into renewable energy development and electricity grid enhancements to withstand climate-related stressors such as heatwaves and floods.

^a Analysis based on ADB Data Library – ADB. [Climate Change Financing at ADB](#).

Energy consumers can play a vital role in promoting private investment in adaptation measures by demanding climate-resilient energy supply. Large corporate buyers—such as technology firms, manufacturers, and hospitals—may negotiate power purchase agreements that include reliability clauses (e.g., 90%–95% uptime guarantees). If energy providers fail to meet these requirements due to climate-driven outages or maintenance issues, large corporate buyers may impose financial penalties. Such contractual arrangements encourage the integration of potential climate-related disruption costs into electricity system planning, ultimately strengthening the reliability of supply under climate stress (Table 5).

Table 5: Actions for Scaling Up Adaptation Finance in the Energy Sector by Stakeholder

STAKEHOLDERS	  		
	National authorities	Energy suppliers	Energy consumers
ACTIONS			
Actions for Scaling Up Adaptation Finance in the Energy Sector by Stakeholder 	• Promote blended finance approaches that combine concessional capital from the public sector with private sector investments • Create an enabling environment for private investment by providing technical assistance for climate risk assessments and demonstrating the viability of innovative technologies • Offer risk-sharing mechanisms through diverse and innovative financial instruments to scale up adaptation finance	• Expand adaptation finance by systematically embedding climate risk assessments in investment planning • Incorporate long-term resilience benefits into decision-making and valuation processes • Leverage insurance as an effective risk-transfer mechanism to manage climate-related uncertainties	• Integrate reliability requirements into power purchase agreements

Source: Author.

Accelerated Deployment of Innovative Adaptation Technologies

Diversifying the energy mix with the increasing penetration of new renewable energy technologies is a long-term solution for climate resilience and energy security. Heavy reliance on a single energy source may limit the capacity of energy systems to adapt to changing climate patterns and extreme weather events. For instance, an energy system dominated by traditional thermal power plants with wet-cooling systems can be critically affected by water shortages caused by droughts. On the other hand, renewable energy technologies such as solar PV and wind turbines, which require significantly less water, can provide alternatives to traditional systems by replacing those thermal power plants.

The deployment of renewable energy not only contributes to technological diversification but also to geographic diversification. Decentralized renewable energy production helps maintain energy supply even when extreme weather events damage one production site. For instance, during the floods in Sabah, Malaysia, in 2021, microgrids with distributed energy sources provided electricity for emergency services amid widespread

power outages.⁶⁹ Given that future climate is likely to increase the level of exposure to multiple climate hazards while placing additional stress on traditional energy systems, building a diverse energy mix is recommended to ensure energy security.

However, renewable energy technologies are not always resilient to climate impacts. In fact, they can be more sensitive to certain types of climate hazards such as extreme heat. For instance, higher temperatures tend to cause lower voltages and reduced electricity generation from solar PV, as solar PV systems are designed to operate best in cool, sunny weather. Solar power generation efficiency degrades as the temperature increases, generally by -0.3% to -0.5% per degree above 25°C. If the surface temperature goes above 35°C and raises a solar panel's temperature to 70°C, solar PV efficiency can drop by 13.5%–22.5%, leading to a notable reduction in generation output if no adaptation measures or technological enhancements are undertaken.⁷⁰ Extreme temperatures can also increase the electrical resistance of the circuits and damage the cells and other materials.

Wind power plants are also susceptible to extreme heat above 40–45°C. Wind power plants are usually designed for a 25°C environment, and a standard wind turbine can work at full power up to an outside temperature of 35°C. Higher temperatures can reduce the life span of battery cells and other electronic components while making the lubrication oil for the turbine gearbox runny, which in turn causes grinding in the gears. Under extreme heat, such as 40–45°C, a standard wind turbine may shut down completely.⁷¹ Given that electricity demand generally soars during extreme heat events, reduced generation efficiency from solar and wind power plants can raise energy security concerns.

To address this issue, innovative designs and technologies are being increasingly adopted (Box 4). Although passive cooling systems based on natural convection and active water-cooling techniques are still being used in many solar PV plants, these methods are increasingly being replaced with more efficient technologies, such as enhanced and combined phase change materials and combined heat pipe and sink cooling. For wind power plants, new designs have been introduced in hot countries to endure 45–50°C. For instance, a new design with enhanced wind turbine ventilation was adopted for the 50-megawatt Dhofar Wind Power Project in Oman so that it could operate in conditions of 45°C.

Digital technologies also lead to climate resilience enhancement in the energy sector, providing tools to monitor, mitigate, respond to and recover from climate-driven disruptions. Technologies such as artificial intelligence (AI), advanced real-time monitoring with satellite imagery and drones, the Internet of Things, as well as machine learning transform the way the energy sector collects, processes and analyzes climate information while improving decision-making against climate risks. Such digital technologies enhance climate resilience of energy systems at different stages of climate-driven disruptions.

Innovative digital technologies improve the accuracy of forecasts of potential outages and enable early warning systems. For instance, Google has developed a Flood Hub, combining two AI models, a hydrological model which forecasts the amount of water flowing in a river, and an inundation model which predicts what areas will be affected and how high the water level will be.⁷² The system is based on publicly available data sources and utilizes satellite imagery to provide riverine flood predictions up to 7 days in advance.⁷³ As most of the countries in South Asia, Southeast Asia, and East Asia are projected to see an increase in riverine floods,

⁶⁹ IEA. 2024. *Climate Resilience for Energy Security in Southeast Asia*.

⁷⁰ Footnote 69.

⁷¹ Footnote 69.

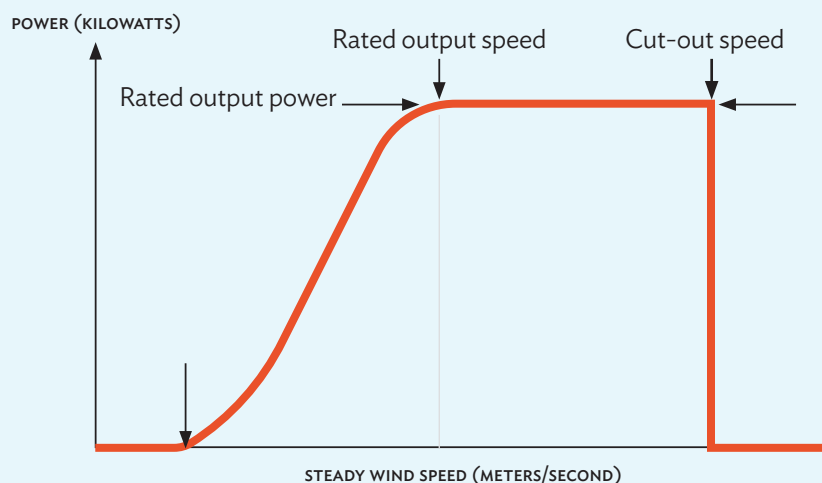
⁷² Google. *Flood Forecasting*.

⁷³ AVPN. 2024. *Digital Tech for Climate Adaptation*.

Box 4: Innovative Wind Turbines Against Intensifying Tropical Cyclones

Wind power plants require certain ranges of wind speeds (approximately 3–4 meters per second [m/s] to 20–25 m/s, with maximum output at about 12–15 m/s) for stable electricity generation. Although wind turbines generally generate more electricity with higher wind speed, an excessive increase in wind speed triggers an automatic shutdown of wind turbines. Wind power plants are set to shut down automatically beyond a certain wind speed limit (around 20–25 m/s) to avoid physical damage. Therefore, more tropical cyclones (above 33 m/s) and severe storms often result in a sharp cut in generation output or even cause physical damage to wind power plants.

Figure: Power Output Variation with Wind Speed



Typical wind turbine power output with steady wind speed

In Asia and the Pacific, changes in wind patterns are likely to pose challenges to the region's ambition to scale up wind power capacity. In recent years, many Asian countries such as India, Indonesia and Viet Nam have committed to rapidly increasing wind power capacities. However, climate models project that wind speed over most Asian regions will decrease while the share of very intense tropical cyclones (category 4–5) is likely to increase.^a The projected decrease in surface wind speed, coupled with an increase in the number of intense tropical cyclones, may threaten wind power generation. Already, some offshore wind plants experienced physical damage from tropical cyclones, as seen in the Honghaiwan Wind Farm, where 17 out of 25 wind turbines were damaged by Typhoon Usagi in 2013.^b

To address the risks of intensifying tropical cyclones and decreasing wind speeds, wind turbine designs and operational practices are increasingly integrating climate resilience. For instance, project sites are selected based on the assessment of wind extremes and blades are strengthened with composite reinforcement. Adaptive pitch control can be used to adjust the angle of the rotor to reduce structural loads.^c Furthermore, storm control modes, such as early curtailment in response to projected tropical cyclones can be implemented.

^a Intergovernmental Panel on Climate Change (IPCC). 2021. *IPCC Sixth Assessment Report: Working Group II: Impacts, Adaptation and Vulnerability*.

^b W. Qi. 2013. *Chinese Typhoon Knocks Out 17 Wind Turbines*. *Windpower Monthly*, 25 September.

^c Q. Yao, et al. 2024. *Anti-tropical Cyclone Load Reduction Control of Wind Turbines Based on Deep Neural Network Yaw Algorithm*. *Applied Energy*. Volume 376, Part B.

enhanced flood forecasts through AI can play a significant role in preventing extreme weather event-related losses.⁷⁴ Google also introduced a FireSat project⁷⁵ aiming to detect and measure wildfires using satellites and AI, which will provide global high-resolution imagery for just 5 meters by 5 meters.⁷⁶ As wildfires can cause disruptions in transmission and distribution networks, such high-resolution detecting tools can help the prevention of faults in electricity grids caused by fires, particularly in countries in Central and West Asia and East Asia, where fire weather is projected to increase. It shows the possibility of applying advanced weather forecasts into energy system management so that it can prepare for potential climate-driven disruptions.

Moreover, digital technologies support faster detection and restoration of faults, including the ones caused by extreme weather events. Drones, satellites, and remote sensors with fixed cameras imagery help energy providers locate and fix faults more quickly, reducing the costs and duration of disruptions.⁷⁷ For example, Viet Nam's National Power Transmission Corporation introduced the AI-based power transmission line management system with the application of uncrewed aerial vehicles for inspection of the transmission grid. It covers over 6,000 kilometers of transmission lines in nine provinces including highlands and coastal areas. It supports more accurate and efficient inspection, drastically saving time needed per inspection and eliminating human errors.⁷⁸ Another example of using innovative technologies for monitoring energy infrastructure can be found in the case of Thailand's state-owned oil and gas company PTT. It enhanced its monitoring capacity via satellite and AI technologies after experiencing a gas pipeline explosion in October 2020. It allowed for more frequent and precise detection of anomalies and changes along the pipelines, helping PTT reduce potential risks and reinforce its resilience.⁷⁹ Such digital technologies for advanced monitoring contribute to improving overall energy system resilience, and support resilient operation against climate-related disruptions.

Energy consumers can support the improvement of the accuracy and applicability of digital technologies by acting as knowledge providers. Energy consumers can report real-time information through social media channels or online applications or even generate preliminary data on energy supply and consumption. In Indonesia, energy consumers are contributing to a tool called Haze Gazer, which provides real-time insights on the location of fire and haze based on multiple data sources, by sharing information and data through social media channels. This information allows the timely identification of fire and haze hot spots.⁸⁰

Energy consumers can also accelerate the deployment of innovative technologies by adopting more efficient options. As warmer temperatures driving energy demand for cooling in Asia and the Pacific, adaptation measures with increased energy efficiency are expected to serve a critical role in climate resilience. In Southeast Asia, where 16% of electricity is currently used in buildings for cooling, the share of cooling in electricity consumption is projected to reach 30% by 2035.⁸¹ As cooling alone will account for almost one-third of Southeast Asia's growth in electricity use, energy efficiency for cooling is becoming increasingly important. In terms of Asia and the Pacific, energy efficiency improvement could reduce annual electricity consumption by up to 40% while behavioral changes in cooling, such as a mix of ventilation, passive cooling, fans and air-conditioning, can also reduce electricity use.⁸²

⁷⁴ Coalition for Disaster Resilient Infrastructure. 2023. *Global Infrastructure Resilience Report*.

⁷⁵ Google. *FireSet*.

⁷⁶ IEA. 2025. *Energy and AI*.

⁷⁷ ADB. 2021. *Digital Technologies for Climate Action, Disaster Resilience, and Environmental Sustainability*.

⁷⁸ EVN Viet Nam Electricity. 2024. *UAV and AI Technology Applications: Breakthrough in Transmission Grid Operation Management*.

⁷⁹ Skywatch. 2022. *How to Use Satellite Imagery to Reduce Risk in Oil & Gas*.

⁸⁰ Footnote 69.

⁸¹ IEA. 2024. *Southeast Asia Energy Outlook*.

⁸² Footnote 19.

Governments can support the adoption of innovative energy-efficient technologies by improving standards and labeling systems. Currently, governments across Asia have different levels of regulations and standards to support efficiency improvement for cooling appliances (Table 6). In Southeast Asia, for example, the Minimum Energy Performance Standards and labeling for efficient air conditioners are still under development in Cambodia, the Lao PDR, and Myanmar while they are already mandated in Indonesia, Malaysia, the Philippines, Thailand, and Viet Nam.⁸³ Enforcing mandatory minimum energy performance based on the latest technologies, along with labeling requirements for appliances, can accelerate the adoption of energy-efficient technologies and provide consumers with essential information about the energy performance.

Table 6: Status of Key Existing Policy Instruments to Improve the Energy Efficiency of Appliances

Country	Buildings		Appliances	
	Building energy codes	Certification/Labeling	MEPS for appliances	Labeling for appliances
Brunei Darussalam	●	● =	● = AC	● = AC
Cambodia	● =	● =	● = AC, R	● = AC, R
Indonesia	● =	● =	● ↑ AC, R, L, F	● = AC, R, L, F
Lao PDR	● =	● =	● = AC	● = AC
Malaysia	● =	● =	● = AC, R, L, F	● ↑ AC, R, L, F
Myanmar	● =	● =	● = AC	● =
Philippines	● ↑	● =	● ↑ AC, R, L, F	● ↑ AC, R, L, F
Singapore	● =	● =	● = AC, R, L	● = AC, R, L
Thailand	● =	● =	● = AC, R, L	● ↑ AC, R, L, F
Viet Nam	● =	● =	● = AC, R, L, F	● ↑ AC, R, L, F

AC air conditioners
R refrigerators / freezers
L lighting
F fans
 = no change since 2022
 ↑ improvements since 2022

Building energy codes, labeling, certification

● Mandatory for all buildings
 ● Mandatory for certain building types
 ● Voluntary
 ● Under development
 ● No known policy

Appliances

Mandatory
 -
 Voluntary
 Under development
 No known policy

Lao PDR = Lao People's Democratic Republic, MEPS = Minimum Energy Performance Standards.

Source: Adapted from International Energy Agency. 2024. *Southeast Asia Energy Outlook 2024*.

⁸³ IEA. 2024. *Southeast Asia Energy Outlook 2024*.

Governments can also encourage behavioral changes by providing incentives. For instance, the PRC encourages local governments to introduce incentives for purchasing high-efficiency cooling appliances through its Green and High Energy Efficiency Cooling Action Plan. Similarly, Japan promotes the use of natural refrigerants as alternatives to hydrofluorocarbons which have high global warming potential. The Ministry of the Environment offers subsidies to cover a portion of the machinery and installation costs for companies that replace or install equipment to use natural refrigerants (Table 7).⁸⁴

Table 7: Actions for Accelerating Deployment of Innovative Adaptation Technologies in the Energy Sector by Stakeholder

STAKEHOLDERS	National authorities	Energy suppliers	Energy consumers
ACTIONS			
Actions for Accelerating Deployment of Innovative Adaptation Technologies in the Energy Sector by Stakeholder 	• Support the adoption of innovative energy-efficient technologies by improving standards and labeling systems • Encourage behavioral changes by providing incentives	• Scale up new renewable energy technologies for diversification of energy sources • Improve the accuracy of forecasts of potential outages and enable early warning systems through digital technologies • Enable faster detection and restoration of climate-related faults with advanced smart technologies	• Enhance the accuracy and applicability of digital technologies as a knowledge provider • Accelerate the deployment of innovative technologies by purchasing more efficient options

Source: Author.

⁸⁴ ADB. 2024. *Sustainable Cooling: How to Cool the World Without Warming the Planet*.

6 Conclusion

Amid escalating impacts due to changes in the climate and extreme weather, the imperative for climate-resilient energy systems has never been greater. The Asia and Pacific region, the most exposed to climate-related hazards in the world, is experiencing increasing climate-driven disruptions in the energy sector. More frequent floods, droughts, and glacier melt threaten power generation from hydropower and thermal power plants while rising sea levels and intensifying tropical cyclones pose severe risks to coastal energy infrastructure, including power plants, fuel production sites, and transmission networks. Furthermore, a warmer climate is putting additional pressure on the energy system rapidly increasing electricity demand for cooling in the region where power supply is already strained due to population growth, economic development and aging infrastructure.

As climate impacts intensify, demand for resilient energy systems is rising sharply. Energy suppliers are under pressure to protect assets and ensure reliable service. Consumers increasingly seek to safeguard their livelihoods and avoid costly interruptions to essential services. Governments and regulators recognize resilience as fundamental to energy security, disaster preparedness, and sustainable development.

Strengthening the resilience of energy systems delivers significant socioeconomic benefits. For suppliers, it means avoiding costly damage, minimizing outages, and ensuring long-term supply chain stability. For consumers, it underpins economic growth, safeguards public services such as health care and water supply, and protects the most vulnerable. For authorities, it advances national energy security, accelerates the clean energy transition, and strengthens disaster readiness. Ultimately, climate-resilient energy systems not only reduce climate-related risks but also generate wide-ranging social and economic dividends that far exceed the cost of action.

Realizing these benefits requires coordinated action. National climate, disaster, and energy authorities play a pivotal role in improving the accessibility and quality of climate information based on climate risk and impact assessments conducted by energy suppliers and feedback from energy consumers. Governments can also lead the initiative to mainstream energy sector resilience into national adaptation strategies and develop a robust implementation plan based on stakeholder consultations. Energy suppliers can support the initiative by embedding resilience considerations into projects and aligning their corporate strategies with national adaptation goals. In addition, energy suppliers are expected to lead investments in resilient infrastructure, while the public sector supports them through blended finance approaches and risk-sharing approaches. The adoption of innovative adaptation technologies can be achieved through collaborative actions among different stakeholders. Energy suppliers can expand deployment of innovative technologies for renewable energy, digitalization, and energy efficiency while energy consumers and governments support them with behavioral changes and relevant policies.

Asia and the Pacific faces both the greatest risks and the greatest opportunities. The region's vulnerabilities are undeniable, but so are the potential gains from decisive adaptation measures. If authorities, suppliers, and consumers would act together, the region can transform climate risk into a catalyst for building robust, reliable, and forward-looking energy systems.

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The Climate-Resilient Energy Imperative

Adaptation Benefits and Actions for All

Climate-resilient energy systems are becoming imperative as the impacts due to the changing climate intensify across the region. Energy infrastructure in the Asia and Pacific region is already under strain, facing power outages due to the impact of changes in the climate, physical damage, and shifting demand patterns. In response to these growing risks, stakeholders are calling for stronger and more coordinated resilience measures. This report outlines the multiple benefits of climate-resilient energy systems, examines current barriers to progress, and proposes targeted actions for key actors—including energy providers, consumers, and government regulators—to accelerate the transition toward a more resilient energy future.

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ADB is a leading multilateral development bank supporting inclusive, resilient, and sustainable growth across Asia and the Pacific. Working with its members and partners to solve complex challenges together, ADB harnesses innovative financial tools and strategic partnerships to transform lives, build quality infrastructure, and safeguard our planet. Founded in 1966, ADB is owned by 69 members—50 from the region.



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